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# Multi-Object Tracking Sensor Fusion: Radar and LiDAR Integration with Kalman Filtering techniques

Beyond Sight: Radar-LiDAR Fusion for Robust MOT

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### Abstract

This thesis presents a multi-object tracking (MOT) sensor fusion framework that integrates radar and LiDAR sensor data using Kalman Filter techniques. The primary objective is to evaluate the complementary strengths of radar within a LiDAR-based framework, Poly-MOT, to enhance the accuracy, robustness, and reliability of MOT in dynamic environments and varying weather conditions. The proposed pipeline employs both the Kalman Filter and the Extended Kalman Filter to optimally combine sensor measurements. Experimental results indicate that while the LiDAR-based framework can operate optimally without the need of a secondary sensor, incorporating radar data helps to improve the tracking performance under adverse weather conditions, providing a more adaptable and robust framework for intelligent transportation applications.

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## Abstrakt

Denna master uppsats presenterar ett multi-object tracking (MOT) sensor-fusion ramverk som integrerar radar och LiDAR-sensordata med hjälp av Kalman filter. Målet med uppsatsen är att utvärdera de kompletterande styrkorna hos radar till ett LiDAR-baserat ramverk, Poly-MOT, för att förbättra noggrannheten och robustheten hos MOT i dynamiska miljöer och varierande väderförhållanden. Den föreslagna processen använder både Kalman filtret och det utökade Kalman filtret för att optimera kombinationen av sensor mätningarna. Resultatet indikerar att även om det LiDAR-baserade ramverket fungerar utan behov av en sekundär radar sensor, så kan integrationen av radardata leda till förbättrad spårningsprestandan under dåliga väderförhållanden, vilket i sin tur bidrar till ett mer robust och pålitligt system.

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## Keywords

Sensor-fusion, LiDAR, radar, data aggregation, Object tracking, Kalman Filter



## Acronyms

**ADAS** Autonomous Driving Assistance Systems. 2, 4, 7–11, 35, 57

**AMOTA** Average Multi Object Tracking Accuracy. 32, 37, 38, 51, 55, 57

**AV** Autonomous Vehicle. 1, 7–12, 14

**BEV** Bird-Eye View. 49, 52

**CNN** Convolutional Neural Network. 35, 37

**DNN** Deep Neural Network. 35

**EKF** Extended Kalman Filter. 20, 21, 35, 36

**FCN** Fully Convolutional Network. 35

**FMCW** Frequency Modulated Continuous Wave. 11

**FN** False Negatives. 29, 32, 33, 38, 44, 51

**FP** False Positives. 32, 51, 56

**IDS** Identity switches. 32, 33, 38, 44, 56

**LiDAR** Light Detection and Ranging. 5, 8–10, 15, 35, 36, 55–57, 62

**mAP** mean Average Precision. 36, 37

**PCD** Point Cloud Data. 8, 36, 37, 60

**radar** Radio Detection and Ranging. 8–11, 36, 62

**SOTA** State Of The Art. 25, 35–37

**TP** True Positive. 32, 44, 51

## Contents

|          |                                                       |          |
|----------|-------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                   | <b>1</b> |
| 1.1      | Autonomous Driving Introduction . . . . .             | 1        |
| 1.1.1    | Vehicle Automation Levels . . . . .                   | 2        |
| 1.1.2    | Key Components of Autonomous Vehicles . . . . .       | 3        |
| 1.1.3    | Localization . . . . .                                | 3        |
| 1.1.4    | Path Planning . . . . .                               | 4        |
| 1.1.5    | Perception . . . . .                                  | 4        |
| 1.1.6    | Current obstacles/issues . . . . .                    | 5        |
| 1.2      | Objectives . . . . .                                  | 7        |
| 1.3      | Authors contributions . . . . .                       | 7        |
| <b>2</b> | <b>Theoretical Background</b>                         | <b>7</b> |
| 2.1      | Sensors in Autonomous Vehicles . . . . .              | 8        |
| 2.1.1    | LiDAR . . . . .                                       | 8        |
| 2.1.2    | RADAR . . . . .                                       | 9        |
| 2.1.3    | Camera . . . . .                                      | 12       |
| 2.1.4    | Sensor summary . . . . .                              | 13       |
| 2.2      | Sensor Fusion . . . . .                               | 14       |
| 2.2.1    | Low-Level Fusion . . . . .                            | 14       |
| 2.2.2    | Mid-Level Fusion . . . . .                            | 15       |
| 2.2.3    | High-Level Fusion . . . . .                           | 15       |
| 2.3      | Object Detection . . . . .                            | 15       |
| 2.4      | Object Tracking . . . . .                             | 17       |
| 2.4.1    | Kalman Filter . . . . .                               | 18       |
| 2.4.2    | Extended Kalman Filter . . . . .                      | 20       |
| 2.4.3    | Filter Tuning . . . . .                               | 22       |
| 2.5      | Data Association . . . . .                            | 23       |
| 2.5.1    | Hungarian Algorithm . . . . .                         | 24       |
| 2.6      | Polyhedral Multi-object Tracking (Poly-MOT) . . . . . | 24       |
| 2.6.1    | Pre-processing . . . . .                              | 25       |
| 2.6.2    | Trajectory Motion Models . . . . .                    | 26       |
| 2.6.3    | Data association . . . . .                            | 29       |
| 2.6.4    | Trajectory Management . . . . .                       | 30       |
| 2.7      | Autonomous Driving Datasets . . . . .                 | 30       |
| 2.7.1    | KITTI . . . . .                                       | 30       |
| 2.7.2    | Waymo . . . . .                                       | 30       |

## CONTENTS

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|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| 2.7.3    | nuScenes . . . . .                                  | 31        |
| 2.7.4    | Evaluation Metrics . . . . .                        | 31        |
| 2.8      | Rigid Body Transformation . . . . .                 | 33        |
| 2.9      | Data Aggregation . . . . .                          | 35        |
| <b>3</b> | <b>Related Work</b>                                 | <b>35</b> |
| <b>4</b> | <b>Methodologies and Methods</b>                    | <b>38</b> |
| 4.1      | Radar Data Integration and Transformation . . . . . | 38        |
| 4.2      | Radar Sensor Fusion . . . . .                       | 41        |
| 4.3      | Kalman Filter and Motion Model . . . . .            | 42        |
| 4.4      | Bad Weather Split . . . . .                         | 43        |
| <b>5</b> | <b>Benchmark and Evaluation</b>                     | <b>44</b> |
| 5.1      | With and Without Velocity . . . . .                 | 44        |
| 5.1.1    | Radar Point Distribution . . . . .                  | 47        |
| <b>6</b> | <b>Experimental Results</b>                         | <b>49</b> |
| 6.1      | Radar Sensor Fusion Results . . . . .               | 50        |
| 6.1.1    | Bad Weather Split . . . . .                         | 54        |
| <b>7</b> | <b>Discussion</b>                                   | <b>55</b> |
| 7.1      | Discussion of Results . . . . .                     | 55        |
| 7.2      | Discussion of Specific Results . . . . .            | 58        |
| 7.3      | Discussion of Method . . . . .                      | 61        |
| <b>8</b> | <b>Conclusion</b>                                   | <b>61</b> |
| <b>9</b> | <b>Future Work</b>                                  | <b>62</b> |

# 1 Introduction

This section will go through the potential of autonomous vehicles (AVs) and the essential components required to bring AVs into reality. It will provide an overview of different levels of autonomous driving, assess where the technology stands at the present and identify current obstacles for AVs. Finally, the section will end with the objective of this thesis and to highlight what the thesis wants address and evaluate.

## 1.1 Autonomous Driving Introduction

Autonomous vehicles are viewed by many as the future of transportation [1, 2]. They have the potential of increasing safety, efficiency and convenience. This potential has contributed to the car industry's constant advancement and high resources put towards AVs. AVs have the opportunity to restructure the transportation of people and goods reducing the number of traffic accidents while simultaneously optimizing the traffic flow. Additionally, AVs have the possibility to be more fuel efficient and thus be better for the environment [3].

According to the World Health Organization there are approximately 1.19 million people each year that die from car accidents and another 20-50 million people that suffer from non-fatal injuries. The group that are most affected are children and young adults ranging from ages 5-29 years [4]. The majority of these accidents are caused by human error such as speeding, distracted drivers and drivers under the influence. Autonomous vehicles have the potential to reduce these numbers by minimizing human errors through advanced algorithms and technology.

AVs also have the opportunity to increase fuel efficiency. A proposal by [3] suggests the idea of a connected system. This approach would enable AVs to optimize routes and coordinate with each other in real-time, leading to smoother traffic flow, reduced congestion, and shorten the time in traffic.

Another possible feature is the capability to help with parking. Once passengers are dropped off, AVs can autonomously seek out suitable parking spots, a feature particularly advantageous in urban environments where parking space is limited.

### 1.1.1 Vehicle Automation Levels

Autonomous vehicles are often called self-driving cars and are vehicles. They are design to operate without human supervision. They instead rely purely on sensors, artificial intelligent, and advanced driver-assistance systems. To better understand the spectrum of autonomy, Society of Automobile Engineers (SAE) has defined six levels of autonomous driving. According to [5], these six levels ranges from 0-5, level 0 being no automation and requires full human control while level 5 is fully autonomous and self-reliant.

Level 0 refers to most vehicles on the road today. These vehicles require humans to control braking, accelerating, and steering. Features at this level can include warnings for blind spots and lane departure.

Level 1 refers to the most basic levels of automation. Features at this level can help with either steering or braking/acceleration. A prime example of this is adaptive cruise control which helps to keep a safe distance from the car in front.

Level 2 refers to partial driving automation and provides features such as ADAS. These systems can help with both steering and braking/accelerating. It is only considered to be partial self-driving since the vehicle still needs to have a human behind the wheel to take over at any time to ensure safety on the road. Tesla's autopilot qualifies as level 2. This level is where most companies struggle to achieve success.

Level 3 refers to conditional driving automation. This level features the ability to both detect and make informed decisions. Level 3 features are capable of detecting a slow car, switch lane and accelerating past it.

Level 4 refers to high driving automation. The key difference to level 3 is that level 4 does not require humans to intervene. The vehicle can solve issues on the road and drive safely on its own. However, this level has its limitations which require the vehicle to operate in certain conditions and areas. An example of level 4 automation is the Waymo taxi service.

Level 5 refers to full driving automation. This level has not yet been achieved but has all the features of level 4 without the area or speed limitation.

### 1.1.2 Key Components of Autonomous Vehicles

To achieve autonomous levels 2 and higher several subsystems need to work and be able to seamlessly synchronise together. Some critical subsystems are localization, path planning and perception. These subsystems account for gathering information of the environment, locate and maintain details and to make decisions to avoid obstacles.

### 1.1.3 Localization

Localization is the subsystem that handles the vehicles position and is often relative to a reference frame. This subsystem is important for the vehicle to be in the correct lane and position itself properly on the road. According to [3], the three most common approaches to achieve accurate localization is by using on-board sensor fusion with Global Position System and Inertial Measurement Unit (GPS-IMU), Simultaneous Localization and Mapping (SLAM) or prior map-based localization.

GPS-IMU fusion approach uses both GPS and IMU sensors in order to strengthen each sensor's weakness. GPS is used to provide global coordinates of the vehicle's position but has issues in urban environments where the satellite signal is weak. IMU helps with this issue, it collect information of the changes in the position and orientation of the vehicle. By combining GPS and IMU measurements they help to create a more accurate and robust localization [6].

SLAM on the other hand, can simultaneously create a map of the environment and track the position of the vehicles. The algorithm is not required to have prior knowledge of the environment and can therefore be used in unknown environments. The issues with this approach is its high computational power and environmental challenges when used outdoors [7].

Prior map-based methods require a detailed information on pre-existing maps. This information often have lane marking, traffic signs and road layouts. It relies on both the pre-existing map and sensor information to be able to make real time decisions. By comparing the data from both sensor and pre-existing map information it is able to make better estimation of vehicles position. This approach is robust and efficient but very costly and depends a lot on the quality of the pre-existing maps. One company that utilizes this approach is Waymo [8].

### 1.1.4 Path Planning

The purpose of the decision-making subsystem is to create a smart driving strategy that prioritizes safety and reliability. To assess this ability, five design criteria are evaluated. These include real-time performance, which is the system's ability to act fast; steering stability; efficient driving and; minimizing the need for emergency braking [9].

For a decision-making system to be reliable, it must have several critical functions. It must have awareness of its surroundings, including the distances and speeds of nearby vehicles and pedestrians. Additionally, it must be able to follow local traffic regulations, which can vary between countries, i.e rules on u-turns, right-of-way, and left and right traffic directions. Furthermore, the system must also have driving ethics, meaning considerations of pedestrian's behaviors and yielding to emergency vehicles i.e police cars, ambulances, and fire trucks [9].

Lastly, the system must exhibit proficiency in both global and local planning. Global planning is creating the optimal route that factors in travel time, energy efficiency, and traffic conditions. Local planning involves making quick decisions regarding acceleration, braking, steering, and lane changes to navigate between obstacles and traffic effectively. By fulfilling these requirements, the decision-making subsystem can contribute significantly to the overall safety and efficiency of autonomous driving systems [3].

### 1.1.5 Perception

Localization and path planning are two crucial subsystems that are needed for Autonomous Driving Assistance Systems (ADAS). However, to create reliable and precise localization and path planning systems, it is imperative to first create a robust and comprehensive perception system. Perception is the cornerstone of autonomous driving and the foundation for both localization and path planning. It provides the necessary information of the surrounding environment that allows localization subsystems to be more accurate in determining the vehicle's position. Additionally, path planning is highly reliant on a well constructed perception system. The vehicle will not be able to make well informed decisions if it has trouble with detecting the objects and traffic on the road.

The perception system's function is to provide the vehicles with information

regarding the surrounding environment, this information include the size and position of objects, free lanes, velocity of objects and prediction of their future trajectories [10]. To extract this information, the perception system often utilizes sensor modalities such as LiDAR, camera and radar. To get a deeper understanding of the perception system, it can be divided into 3 components, semantic segmentation, object detection and object tracking.

Segmentation algorithms cluster the point clouds and pixels into groups which are then passed through the classification stage that assigns these clusters with labels of different categories. Class labels are useful to recognize objects and help machine learning models to learn different object characteristics [3].

The purpose of object detection is to identify the size, velocity and position of dynamic objects such as vehicles, bicycles and pedestrians, and static objects like road signs and traffic lights [3]. Object detection algorithms can utilize sensor fusion techniques to strengthen one sensor's weakness with data from another sensor. Sensor fusion has the possibility to make the object detection algorithm more robust and effective [10].

Object tracking algorithms purpose is to predict future trajectory of objects to avoid collision [3]. Object tracking algorithm commonly uses LiDAR sensors that provide the AV with range estimation and there are tracking algorithms like CenterPoint [11] that also utilizes LiDAR to estimate the velocity by taking the object's position from two timesteps divided by the time difference. Additionally radar sensors can also be used in object tracking to provide velocity estimation. Typically, the information extracted from these sensors are utilized to estimate the positions, headings, and velocities in conjunction with a Kalman filter or deep learning models to predict future trajectories. This will be further discussed in the 2.4 section.

### 1.1.6 Current obstacles/issues

At the present most autonomous vehicles operate at level 2-3 of autonomous driving, where there is still a need for a human to operate and intervene when necessary. However, there are companies like Waymo taxi services, that operate at level 4 autonomy, where vehicles can operate without a human operator.

Waymo is able to operate at level 4, largely due to their pre-mapping so-



lution which is a method that creates a very detailed map of the road and surrounding environment before the actual drive. It allows the vehicles to use pre-existing knowledge to become more accurate and efficient. However, the issue with this solution is that it is very costly to create detailed map of a large area and faces issues in changing environments where there is a lot of construction [12]. Another issue with this method is the scalability factor. Pre-mapping is a resource heavy method and will require high computation power and storage capabilities when creating a detailed map for thousands and thousands of kilometers.

Another challenge for autonomous driving is the limitations of current sensor technology. As previously mentioned the most common sensors for perception are LiDAR, radar, and cameras. LiDAR is a sensor that excels in medium range, 3D mapping, high precision in distance measurement and suitable for object detection and tracking. However, it has challenges with high surface reflection, high cost and struggle in harsh weather conditions. Radar is a sensor that can provide at velocity measurement, long range detection and high tolerance to weather interference, but subsequently has low resolution, sparse points clouds, and high noise. Camera offers color information and is good at classification with a wide range of view. Downside to cameras they require high processing speed and are bad in low lights and harsh weather conditions such as snow and rain.

A solution to the limitations of each sensor is sensor fusion. The theoretical idea is to combine the strengths of each sensor. An obstacle for sensor fusion is that the sensors in question are drastically different and fusing the information of these sensors is nontrivial. It also requires high computational power and a complex model that can affect latency [13]. This specific task is challenging given the big ammount of data collected by the sensors and pre-processing required for sensors.

To conclude, sensor technology needs to be enhanced, they need to be more accurate, have higher resolution and be more robust to provide perception systems with more details. These issues must be addressed to reach level 5, fully autonomous vehicles.

### 1.2 Objectives

The objective of this thesis is to integrate radar sensors through sensor fusion techniques with Kalman filter on an existing framework utilizing nuScenes dataset to analyze the impact of radar for object tracking. The thesis aims to determine in which scenarios radar is especially good and whether the current radar sensor is beneficial enough to justify the computational overhead.

This thesis further seeks to explore the current obstacles faced by adverse weather conditions, specifically in object tracking. Additionally, propose solutions regarding how to make a robust ADAS system in adverse weather conditions such as heavy rain, fog and snow.

### 1.3 Authors contributions

**Henrik:** 1 Introduction, 2.3 Object Detection, 2.4 Object Tracking, 2.5 Data Association, 2.6 Poly-MOT, 2.7 Autonomous Driving Dataset, 3 Related Work, 4.2 radar Sensor Fusion, 4.3 Kalman Filter and Motion Model, 5 Benchmark and Evaluation, 6.1 radar Sensor Fusion Results, 7.1 Discussion of Results, 7.2 Discussion of Specific Results, 7.3 Discussion of Method, 8 Conclusion

**Kevin:** 2.1 Sensors in Autonomous Vehicles, 2.2 Sensor Fusion, 2.4.1 Kalman Filter, 2.4.2 Extended Kalman Filter, 2.4.3 Filter Tuning, 2.5.1 Hungarian Algorithm, 2.5.2 Greedy Algorithm, 2.8 Rigid Body Transformation, 2.9 Data Aggregation, 6.1.1 Bad Weather Split, 7.1 Discussion of Results, 9 Future Work

## 2 Theoretical Background

This section will cover the theory behind the multiple parts associated with AVs. It will cover the most commonly used sensors in AVs, the different types of sensor fusion and the upside to sensor fusions. Additionally, it will cover the theory behind object detection, object tracking and Kalman filters. This section will further present the dataset and evaluation metric used for this thesis, and include the details of Poly-MOT which is the tracking framework used as the foundation of this thesis. Understanding the theory behind these components will make it easier to understand the result and discussion sections.

### 2.1 Sensors in Autonomous Vehicles

Sensors have a main role in enabling AVs to perceive the surroundings, ensuring safe and efficient travel. Exteroceptive sensors, also known as external state sensors, gather external data from the vehicle's environment, including object distance, light intensity, and positional data. Key sensors within this category are cameras, LiDAR and radar sensors which all have a crucial role in ADAS. Examples for such sensors can be found in Table 1. Sensor data makes it possible for the AV to make real-time decisions, avoid collisions, and drive safely on the road [14].

#### 2.1.1 LiDAR

Light Detection and Ranging sensors works by emitting laser light and receiving the reflection to measure the distance with high precision and resolution, enabling the mapping of the surrounding environment. The LiDAR sensor is composed of a laser scanner that emits infrared light and a detector that collects the returning light pulses. The sensor calculates the distance to objects by measuring the time it takes for light pulses to travel to an object and reflect back to the sensor. Since the speed of the light is a constant, precise distance calculations can be done. These calculations provides a dataset containing the precise time and distance of surrounding objects. The output data results in a point cloud data (PCD), able to position objects on the 3D plane  $(x, y, z)$  and provides intensity information of the objects ( $i$ ) that represents the reflectivity of each data point, indicating how much light is captured by the sensor. LiDARs rapid emission of light pulses gives an accurate description of the surrounding objects while generating an abundant dataset. Every hour of AVs operation can create up to 1 terabyte of data, creating a barrier in the vehicles data storage capacity. Consequently, numerous processing steps to manage the abundant data are required, including downsampling the PCD or cleaning noisy data [15, 16].

There are two main types of LiDAR sensors, the airborne LiDAR and the terrestrial LiDAR. Airborne LiDAR systems are used to create 3D models of the earth surface to e.g. analyze forest canopy or topographic surface models. The terrestrial LiDAR are used to localize and map surrounding objects with high precision and are the ones used for this thesis. There are various types of terrestrial LiDAR:

**Mechanical LiDAR** sensors has the ability to rotate at  $360^\circ$ , scanning the surroundings in all directions to form a point cloud. These sensors use mirrors to direct the laser pulses in a specific direction, allowing an accurate mapping of the environment [15]. Widely used in AVs, mechanical LiDAR is a traditional solution for perception systems. However, its components are large in size and heavy, and the rotating mechanism is prone to failure due to mechanical stress. Additionally, compared to the their solid-state LiDAR counterparts, these sensors have higher cost and limited scanning speed, resulting in a relatively lower data acquisition rate [17].

**Solid-State LiDAR** sensors employs stationary components, using a solid-state laser and photodetectors to emit and detect light pulses without any moving parts [15]. This stationary design increases the sensors durability. In contrast to the mechanical LiDARs, solid-state LiDAR are compact and lightweight, expanding their utility across various applications such as drones and AVs. Without the need for rotation, these sensors achieve higher data acquisition rate, enabling accurate mapping of high-resolutions within a single area, useful for obstacle detection and navigation [17, 18]. While stationary components offer many advantages, they also limit the sensor's ability to achieve a  $360^\circ$  field of view and can result in a shorter detection distance, often requiring multiple devices to operate effectively.

Although LiDAR-sensors are known for their accuracy and precision, they face challenges during harsh-weather conditions, resulting in decreased accuracy levels of up to 25% [14, 19]. Addressing this issue is crucial for the future of LiDAR technology. Future challenges include increasing the performance while simultaneously enhancing affordability and reducing the size and weight of the sensors.

### 2.1.2 RADAR

The development of radar, which started during the 1930s, played a crucial role in military defense during the World War II. By analyzing the reflections of electromagnetic waves from targets, radar provided vital information about the position and speed of hostile aircraft, making itself an essential tool in military operations [20].

In modern times, radar technology has extended beyond military use to include automotive systems, particularly in ADAS. In these systems, radar

sensors has a crucial role for detecting the velocity and position of objects, even during challenging weather conditions, thereby enhancing the reliability of autonomous driving.

One of the primary applications of radar is based on the Doppler property of electromagnetic waves [14]. The Doppler effect occurs when there is motion between the source of a wave relative to the observer, causing a shift in the frequency received by the observer [21]. This shift is proportional to the relative velocity between the source and the observer. The Doppler effect follows these equation:

$$|f_D| = \frac{2v \cos(\theta)}{\lambda} = \frac{2v \cos(\theta) f_{tx}}{c_0}, \quad (1)$$

$$v_r = v \cos(\theta), \quad (2)$$

where  $f_{tx}$  is the transmitter frequency,  $c_0$  the speed of light, and  $f_D$  the Doppler frequency shift, representing the difference between the transmitted and received frequencies. This shift is essential for calculating the radial velocity  $v_r$ , which is the component of the target's velocity  $v$  in the direction of the radar beam, determined by the angle  $\theta$  between the target's direction of motion and the radar beam, as represented in [22].

In the automotive sector, during the advancement of ADAS, radar sensors emerged as key component to be mounted on ground vehicles. They play a crucial role in accurately perceiving the velocity of objects, while also having a crucial impact of collision avoidance, as well as the detection of vulnerable road users (VRUs) and static obstacles [23]. This feature has lead to the widespread application of radar sensors in autonomous emergency breaking (AEB) systems for AVs. One important attribute of these sensors is the frequency of the electromagnetic waves, these waves are longer making radar unaffected by lightning conditions or weather conditions [24]. This feature not only enhances safety by properly functioning across various lightning conditions but also performs well during rain, snow or fog. Moreover, radar sensors possess the capability of effectively distinguishing between stationary and moving objects while simultaneously detecting multiple objects at the same time. However, radar faces challenges in accurately detecting smaller objects due to its lower resolution range compared to LiDAR [24]. This

limitation affects the detection and classification performance of surrounding objects. Additionally, radar sensors are susceptible to interference from external sources, further affecting the performance. While radar serves as a key sensor in the ADAS, it can not operate alone to provide an AV with sufficient information for safe navigation [24, 25].

There are many types of radar sensors categorized based on their range that play crucial roles in different functions such as parking, collision avoidance and cruise control systems [26].

**Frequency Modulated Continuous Wave (FMCW)** Radars are one of the most common sensor in the AVs industry. This radar system operates by continuously transmitting a frequency that varies over time at a known rate. This systems measures the frequency difference,  $\Delta f$ , between then transmitted and received signals. Having the advantage of low-cost digital processing requirements, FMCW radars transmits continuously signals to extract information on distance, velocity and angle of surrounding objects [23]. FMCW uses separate receiving and transmitting antennas to avoid signal leakage. The main features for this sensor include assistance-, guidance- and safety-systems. Compared to other radar sensors, FMCW displays an advantage in cost, signal processing and a system simplicity [27, 28].

**Short-Range Radar (SRR)** has an frequency operation range within 24-77GHz of up to 30m. The main functions include parking assistance systems and blind spot detection while providing accurate distance measurements. These functionalities require precise object movement tracking for the prediction of trajectories, a task SRR's high resolution is well-suited for [29, 30, 31].

**Medium-Range Radar (MRR)** functions at 77GHz range of up to 150m. This sensor is mainly utilized in adaptive cruise control systems, where it detects and tracks vehicles to assist maintaining a safe distance [29, 31].

**Long-Range Rada (LRR)** has a frequency operating range from 77GHz of up to 250m, surpassing the capabilities of SRR and MRR. This extended detection range enables the sensor to detect objects at distances of hundreds of meters, resulting enhanced awareness thus improving collision avoidance and safety [29, 32].

**4D-Radar** Similar to the previously mentioned radar sensors, the 4D radar emits radio waves that bounce off objects and return to the sensor, utilizing

Doppler effect to calculate position and velocity. Advanced radar systems, often marketed as 4D radar, offer enhanced resolution, including elevation information. This is achieved through sophisticated antenna placement, advanced beam reflection techniques, and complex signal processing algorithms [33].

### 2.1.3 Camera

The camera is one of the primary sensors used for object classification, scene perception, and color perception tasks. Cameras work by capturing the light emitting from objects which provides a picture in 2D. The emitting light has different wavelengths that represent various colors, for the camera to be able to understand each color it is needed a Red Green Blue- (RGB) camera. Filters inside the lens allow specific wavelengths of light to pass through, enabling the camera to separate and identify different colors more accurately [14].

Cameras stand as the predominant sensors in the field of automotive perception due to their affordability and great classification properties, making it an ideal modality to use on AVs [14]. On the other side, camera weakness lies in the loss of depth information that occurs when the data points are visualized in a 2D space and has challenges during bad weather or poor lighting conditions.

#### 2.1.4 Sensor summary

As shown in Table 1, the various sensor modalities shine in different areas. LiDAR generally excels in range, resolution, and accuracy but is costly and power-demanding. Radar performs well in all-weather conditions and has good range and detection capability but is not able to provide good resolution quality. Cameras offer high resolution and color information at a lower cost and power consumption but are less effective in poor weather conditions.

| Characteristics       | LiDAR | Radar | Camera |
|-----------------------|-------|-------|--------|
| Range                 | +     | ++    | +      |
| Resolution            | +     | -     | ++     |
| Accuracy              | +     | -     | +      |
| Cost                  | -     | +     | +      |
| Weather Tolerance     | -     | +     | -      |
| Object Classification | +     | -     | +      |
| Detection Capability  | +     | -     | +      |
| Processing Load       | +     | ++    | +      |
| Data Density          | +     | -     | +      |

Table 1: Comparison of LiDAR, Radar, and Camera Sensors Across Various Characteristics.



## 2.2 Sensor Fusion

As previously mentioned, AV systems rely on the integration and coordination of sensor data to enhance the situational awareness. Each sensor in the system offers unique advantages, as illustrated in Table 1. Sensor fusion combines data from these individual sensors, allowing the system to overcome independent limitations while enhancing the AV’s safety, performance, and detection accuracy, see Figure 1.

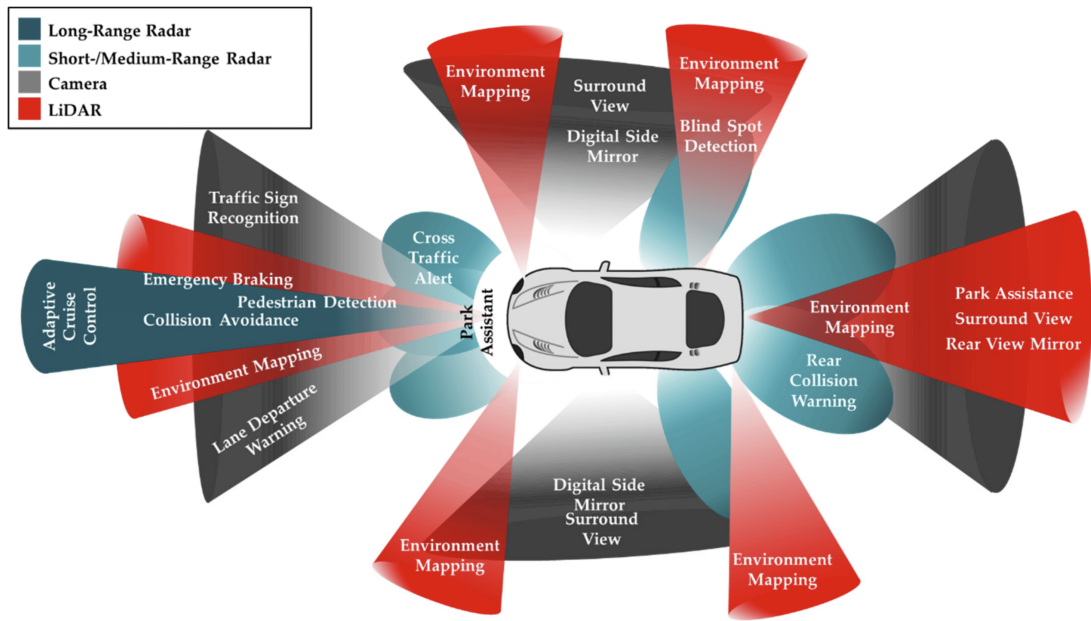


Figure 1: Various sensor modalities strategically positioned on an automated vehicle with different ranges and coverage capabilities, extracted from [14].

### 2.2.1 Low-Level Fusion

At the lowest level of abstraction, known as Low-level or Data-level fusion, raw data from multiple sensors are combined. This method uses all data, enhancing accuracy above the individual original sources. This level key function includes feature extraction, data classification, and object detection [14]. While this approach can provide great results, it requires precise calibration between the modalities for synchronizing data from multiple sensors and requires high processing power. An example of sensor fusion at this ab-

straction level is the fusion of lidar-camera data, where 3D raw point cloud data from LiDAR is fused with 2D images from the camera. This method allows projection of the point cloud on the image leading to a 2D object detection [34].

### 2.2.2 Mid-Level Fusion

Mid-level fusion, also known as Feature-level fusion, is an abstraction level between low- and high-level fusion. At this level, various features are extracted from each sensor modality and fused to build a high-dimensional model that combines these features. The content of each feature includes diverse information, such as color data from camera sensors or spatial information from LiDAR or radar sensors [14]. This method is commonly used to extract the most valuable features from each modality, improving recognition accuracy. Extracting and identifying significant features in e.g. machine learning requires large training sets, which can heavily be affected by inaccuracies in sensor data calibration [34].

### 2.2.3 High-Level Fusion

In this approach, each sensor performs an object detection or tracking algorithm which is subsequently fused into one common algorithm. By fusing the processed information, we achieve a model with lower model complexity compared to other fusion levels. However, relying on processed data implies that valuable information gets lost, reducing confidence and potentially decreasing model accuracy [14, 34].

## 2.3 Object Detection

Object detection is used for multiple applications such as security systems, diagnosing diseases and in autonomous vehicles. Within autonomous vehicles it leverages sensors such as LiDAR, radar and camera to detect, classify and locate objects in the nearby surroundings. Object detection's main goal is to provide useful information of the vehicle's surroundings to be able to plan the road ahead, navigate and avoid obstacles in traffic [2]. For this thesis

The current state of the art object detection models can, according to [35] be divided into two-stage and one-stage detectors. Two stage detectors use the first stage to create Region of Interest (RoI) on the current frame which

is a value of how interesting an area is. The RoI values are often determined based on various features such as texture, color, and edges that suggest the presence of an object. The second stage involves selecting RoIs that meet a specific threshold value then creating bounding boxes for the identified object in that area. One stage detectors, on the other hand, creates bounding boxes directly on the current frame. The advantages of two-stage detectors are in their accuracy, but at the cost of speed. Examples of such frameworks include R-CNN and Fast R-CNN [36],[37]. On the other hand, single-stage detectors offer faster processing and simpler architectures, facilitating end-to-end learning. Popular frameworks in this category include CenterPoint, and VoxelNet [11, 38].

Expanding the scope of object detection, models can also be categorized as either 2D detectors or 3D detectors. 2D detection models often utilize cameras and sometimes radar to provide the horizontal and vertical position of detected objects, but lack the information of depth which is essential to calculating size, shape and position. In contrast, 3D detectors can provide information in x, y, z coordinates, heading angle, width, length and height of detected objects. 3D detection commonly utilizes LiDAR sensors, and more complex frameworks use sensor fusion techniques to incorporate camera and radar sensors to provide a more accurate information [35].

These models require a lot of data in order to learn the complex patterns in dynamic traffic. They are used to extract meaningful features such as color and shape and to achieve an robust models a the training dataset needs to be large but also consist of various environments and weather. Most deep learning frameworks use bonding boxes to locate and detect objects in the vehicle's field of view extracted from LiDAR sensor[39].

After a deep learning model is properly trained their primary application is to classify and locate objects with bounding boxes. Bounding boxes are rectangular shapes that enclose a detected object. These boxes have coordinates that make it easier for the control system to understand the size and distance between the vehicle and the identified objects. The bounding box information is vital for maneuvering and for navigation in a dynamic environment [39].

Although there has been a lot of advancement within object detection in recent years there are still a lot of challenges left. There are fundamental limitations from sensors such as diffusion, blurring, and occlusion that comes

from bad weather and difficult lighting conditions that affects the accuracy and reliability of object detection. There are also practical limitations such as computer resources and latency issues. Object detection models require large input data and can be very complex. This can impact the real-time decisions and hinder the ability to make decisions quickly and thus having a negative impact on safety [40].

One solution is to improve the technology for each sensor to become more accurate and robust. The advancements in LiDAR sensor technology need to continue to improve on tackling the sensor's performance adverse weather in order to become more robust. Also achieving a higher range to detect objects at greater distance to provide more information and time for the vehicle to respond. There should also be efforts in lowering the cost of LiDAR sensor to make them more accessible.

Radar sensor technology also needs advancements specifically with their accuracy and resolution in order to effectively monitor the speed and position of multiple object on the road and facilitate more accurate predictions of the future state of an object and improve object tracking. High-density radar such a 4D radar is a good example of this. 4D radars offer much higher resolution and an another dimension, providing elevation. This improves the radar sensor to detect more objects and have the potential of improving object detection and tracking.

### 2.4 Object Tracking

Object tracking is another important ability for an autonomous vehicle to be able to navigate safely. The state of the art object tracking algorithms commonly use a tracking-by-detection paradigm meaning that the algorithm first detects an object with bounding boxes in each frame and then applies object tracking. Unlike object detection, tracking involves continuous location and monitoring of objects in each frame. Object tracking is used for several objectives, it must be able to know and keep track of the multiple identities and follow them respectively. Ensuring the ability to track the same objects that leave and reenter the frame. Another aspect of the object tracking algorithm is its ability to estimate and predict the object's trajectory to further anticipate the object movement and plan accordingly [41].

Object tracking can be divided into several crucial stages that enable it to

accurately monitor and predict an object movement. The first stage begins with pre-processing the information from the bounding boxes used for object detection. This stage decides which bounding boxes to track by only including bounding boxes with a score higher than a certain threshold [42].

The second stage is the motion model, where classification holds significant importance. Achieving precise classification of pedestrians, vehicles, and bicycles is important due to the distinct movement characteristics from each object. Accurate labeling is crucial in predicting the motion of the trajectories, thereby enhancing overall safety and efficiency. Kalman filters are often ingratiated with the motion model to generate predictions based on a specified state-transition model, which includes assumptions about how the system evolves over time. It then compares these predictions to actual measurements, allowing it to iteratively refine its estimates with each iteration. Following the motion model is data association, there are several data association algorithms such as Hungarian algorithm and Greedy algorithm. These algorithms are inherently different but their premise is the same, pairing the detected object with a trajectory by matching feature similarity or the spatial proximity, this will be discussed more thoroughly in Section 2.5.1. Lastly is Life cycle, that determines if an object should be tracked “birth” and also when an identified object no longer need to be tracked and should be “killed” [43].

Similar to object detection, object tracking also has several challenges in complex environments, occlusions, and varying lighting conditions. A possible solution for addressing this issue is with sensor fusion, which will be further explored in this thesis.

### 2.4.1 Kalman Filter

The Kalman filter is a powerful mathematical technique able to estimate and control the state of a dynamic system over time, even when the information about the system’s state is incomplete. This statistical method was pioneered by Dr. Rudolf Kalman in the late 1950s and detailed in his scientific paper titled “A New Approach to Linear Filtering and Prediction Problems” [44].

The Kalman filter has had a significant impact on society, as it enables the solution of complex problems that were previously unsolvable. In the

early 1960s, the Kalman filter demonstrated its great utility in aerospace and military applications, particularly in guidance, navigation, and control systems. Even today, the Kalman filter continues to show a great range of applications such as radar tracking and time-series analysis.

The Kalman filter consists of a series of dynamic equations that are divided into two key stages: prediction and update, which form optimal estimators for linear systems. This model is highly effective in processing sequential measurements, from e.g. sensors, to generate an estimate of the system's quantities, making it particularly useful for sensor fusion applications.

The Kalman filter is constructed from a linear state space model

$$x_{k+1} = F_k x_k + E_k + w_k \quad (3)$$

$$z_k = H_k x_k + v_k, \quad (4)$$

where  $k$  represents the current time,  $x_k$  the state vector,  $F_k$  the state transition matrix,  $E_k$  the control of the inputs,  $z_k$  the observation measurement, and  $H_k$  the observation matrix. The filter noises are given by the process noise  $w_k$  and the measurement noise  $v_k$ .

In the case of this thesis, the state vector  $x_k$  can contain information about the position and velocity for an object, and the observation state  $z_k$  the measurements from the LiDAR and/or the radar sensor, see Section 4 for more detailed information.

The goal is to create an estimation of the state  $x_k$  from the observation state  $z_k$ . Assuming  $x_k$  and  $z_k$  follow (3) and (4), with  $w_k$  and  $v_k$  being independent white noise processes with distributions  $N(0, R_k)$  and  $N(0, Q_k)$  respectively. The optimal estimate  $\hat{x}_k$  follows the Kalman filter which consists of the prediction step

$$\hat{x}_{k|k-1} = F_{k-1} \hat{x}_{k-1|k-1} + E_{k-1} \quad (5)$$

$$P_{k|k-1} = F_{k-1} P_{k-1|k-1} F_{k-1}^\top + Q_{k-1} \quad (6)$$

and the update step

$$\tilde{y}_k = z_k - H_k \hat{x}_{k|k-1} \quad (7)$$

$$S_k = H_k P_{k|k-1} H_k^\top + R_k \quad (8)$$

$$K_k = P_{k|k-1} H_k^\top S_k^{-1} \quad (9)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_k \quad (10)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1}, \quad (11)$$

where  $\hat{x}_{k|k-1}$  is the estimate of the state  $x_k$  using measurement until time  $k - 1$  and  $P_{k|k-1}$  is the corresponding prediction error covariance.

It is important to note that the noise covariance matrices  $R_k$  and  $Q_k$  vary across models. These matrices enable precise tuning to reflect the specific noise characteristics affecting each parameter. Properly tuning these matrices ensures that the filter accurately mimics real-life dynamics and enhances the Kalman filter's performance. See Section 2.4.3 for a more detailed explanation.

By utilizing the innovation covariance  $S_k$ , the filter can calculate the Kalman gain  $K_k$ , which along with the innovation equation  $\tilde{y}_k$ , is able to calculate the *a posteriori* state estimate  $\hat{x}_{k|k}$ .

The effectiveness of the Kalman filter relies on certain requirements and limitations. Specifically, its optimal application is for linear processes with noise that follows a Gaussian distribution [44, 45].

### 2.4.2 Extended Kalman Filter

The Extended Kalman Filter EKF extends the original Kalman filter discussed in 2.4.1 to handle non-linear systems by locally linearizing around the current estimate [46]. Unlike the Kalman Filter, the EKF can approximate the behaviour of non-linear systems, making it highly versatile and useful in a wide range of applications such as navigation and sensor fusion [47].

In this case, the state vector  $x_k$  and the observation model  $z_k$  are defined by the two equations

$$\mathbf{x}_k = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k \quad (12)$$

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad (13)$$

which are able to handle non-linear dynamic equation. Since these equations describe non-linear dynamics, the standard Kalman filter cannot be applied directly. By linearizing (12) and (13) around the current state estimates at each time step, the EKF can iteratively predict and update the state estimate and covariance, enabling the filter to effectively manage the non-linearities

present in the system. The extended Kalman filter relies on the differentiable functions  $f$  and  $h$  instead of the linear functions  $F_k$  and  $H_k$ .

The EKF, similar to the standard Kalman Filter, consists of two computational steps i.e. prediction step and the update step.

### Predict

As presented in [48], during the prediction step the filter predicts the future state of the system based on the current state measurement given by

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_k) \quad (14)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^\top + \mathbf{Q}_k \quad (15)$$

$$\mathbf{F}_k = \left. \frac{\partial f(\mathbf{x}, \mathbf{u}_k)}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1|k-1}} \quad (16)$$

showing a similar process as the standard Kalman filter, with the key difference being the use of the Jacobian matrix  $F$  that locally linearizes the non-linear state transition function around the current state estimate. This local linearization enables the EKF to handle and approximate non-linear systems effectively.

### Update

In the update step the EKF corrects the predicted state estimate using the apriori measurements given by

$$\hat{\mathbf{z}}_k = h(\hat{\mathbf{x}}_{k|k-1}) \quad (17)$$

$$\mathbf{y}_k = \mathbf{z}_k - \hat{\mathbf{z}}_k \quad (18)$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k \quad (19)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1} \quad (20)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad (21)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (22)$$



$$\mathbf{H}_k = \left. \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}_{k|k-1}} \quad (23)$$

During the update step, the measurement residual  $y_k$  tells the difference between the actual observation and the predicted value. This information is then used to find out the uncertainty of the data, the smaller the uncertainty the more certain the EKF is about the predicted measurement.

The state covariance matrix  $P_k$ , estimates the uncertainty associated with the predicted state. This information is crucial for calculating the Kalman gain, which determines how reliable each measurement is.

### 2.4.3 Filter Tuning

To ensure the optimal operation of both the Linear Kalman filter and the Extended Kalman filter, achieving statistical consistency is crucial. This requires the filter to describe the state error for any given data. Achieving statistical consistency requires tuning the filter to mimic the ground truth data. Filter tuning, as discussed in the literature on Bayesian Optimization [49], involves hyperparameters adjustment to optimize the filter's performance. This tuning process is based on the Kalman filter properties: (i) the filter's ability to assess its potential error, and; (ii) its capacity to blend information from both the process and measurement models to correct prediction estimates effectively.

The hyperparameters in question are process noise covariance matrix  $Q_k$  and measurement noise covariance matrix  $R_k$ . The uncertainty in the dynamic model, reflecting how accurate the model is and how much it deviates from the predicted state is described by  $Q_k$ . This covariance noise matrix is added into the state vector  $x_k$  to take account for the uncertainty in the predicted state, seen in (6) and (15) for KF and EKF respectively. Increasing the process noise causes the filter to place more trust in the measurements than in the model predictions, leading to stronger reaction to each new measurement.

The covariance noise matrix  $R_k$ , shown in (8) and (19), describes the trust on the sensor measurements, indicating the noise level or reliability of the sensor. This noise covariance affects the observation model  $z_k$ , determining the weighting between the predicted state and the new measurements. Increasing the measurement noise covariance reduces the filter's reliance on the

sensor measurements, relying more on the prediction values. On the other side, decreasing this parameter makes the filter trust the sensor data more, assuming it to be accurate. The hyperparameters, in other words, represent the trust factor the system has on the measurement and process values.

Since the noise covariance  $Q_k$  and  $R_k$  are matrices, it is possible to adjust the noise level for each parameter individually, allowing precise modeling of the noise characteristics in the filter. However, this covariance adjustment can be challenging. The general approach is to manually tune these parameters until the filter performs as expected. Manual tuning can be done by choosing values that estimates as accurate as possible the ground truth data, a method known as heuristic tuning. To speed up this process, certain assumptions can be made. For instance, if the sensor data is known to be inaccurate, the measurement noise covariance  $R_k$  can be increased to reflect this [50].

### 2.5 Data Association

Data association is a critical part of object tracking, with data association the premise is to match the correct object detection with its corresponding trajectory. As previously mentioned in 2.4 this thesis is working with a tracking-by-detection paradigm, meaning that the first frame will pass through a detection model followed by an object tracking model. For the first frame it's typical for object tracking models to assign every detected object with a tracking ID and then predict the object movement to the subsequent frame. These predictions are often generate from Kalman filter and the chosen motion model and will continually be updated and associated with the detected objects in the following frame. If the Kalman filter generates an inaccurate prediction, it may result in a lost trajectory, where it fails to correctly associate the trajectory with the corresponding object. A suitable data association metric to solve this issue is the Hungarian algorithm, which optimally pairs predicted trajectories with detected objects, ensuring accurate tracking, see 2.5.1 for further explanation.

To illustrate, consider a scenario where two pedestrians walk closely together in a video frame. Each pedestrian is detected with a bounding box in the current frame (Frame 1). The tracking system predicts their trajectories for the next frame (Frame 2) based on their current positions and velocities. In Frame 2, due to their proximity and similar trajectories, the system assigns predicted trajectories (predicted bounding boxes) to each of actual detected

pedestrian bounding boxes in Frame 2. Now, the challenge is to correctly associate each detected bounding box from Frame 2 with the corresponding predicted trajectory. Algorithms like the Hungarian algorithm or Greedy algorithm are then used to associate each detected bounding box with its best-matched predicted trajectory based on similarity metrics. Data association of detected object and trajectories ensures that object tracking maintains the same tracking ID across frames, enabling robust and continuous tracking in dynamic environments.

### 2.5.1 Hungarian Algorithm

The Hungarian Algorithm introduced by Kuhn [51] can be used to solve the assignment problem between optimal detection-tracking pairs in the number of targets by minimizing the total assignment cost of each target.

The cost matrix is a fundamental component of the Hungarian Algorithm. This matrix is a square-matrix where each element has a 'penalty' or so called *cost* associated with each particular detection to a specific target [52]. In the tracking task, the cost value can be determined by factors as distance between predicted and actual position points, velocity difference or various other metrics that represent the likelihood of true assignment.

The rows of the cost matrix represent the detections, while the columns represent the radar sensor information about each object on the road. In other words, each column corresponds to a data point containing information about a detected object. The Hungarian Algorithm is known for its efficiency and polynomial-time complexity  $O(n^3)$ . It operates by minimizing the total cost represented by the values in the cost matrix. The algorithm's steps involve row/column reduction, zero assignment and adjustments of the cost matrix to achieve optimal detection-tracker pairs [53].

During this thesis this method will be used to accurately associate the most suitable point of data containing radar information together with their corresponding detected objects.

## 2.6 Polyhedral Multi-object Tracking (Poly-MOT)

For this thesis the Poly-MOT framework has been used to develop a robust object tracking system through sensor fusion by integrating radar sensors. Poly-MOT is a 3D Multi-Object Tracking algorithm that contains 4

## 2 THEORETICAL BACKGROUND

key parts, preprocessing module, multi-category trajectory motion module, multi-category data association module, and trajectory management module see Figure 2 for the overview of the framework [54].

What separates Poly-MOT from other tracking frameworks is that it takes the object class into consideration and have different motion models for different class objects both in the motion module and association module. This framework is established as State Of The Art (SOTA) object tracking algorithm and ranks nr. 2 on nuScenes object tracking leaderboard.

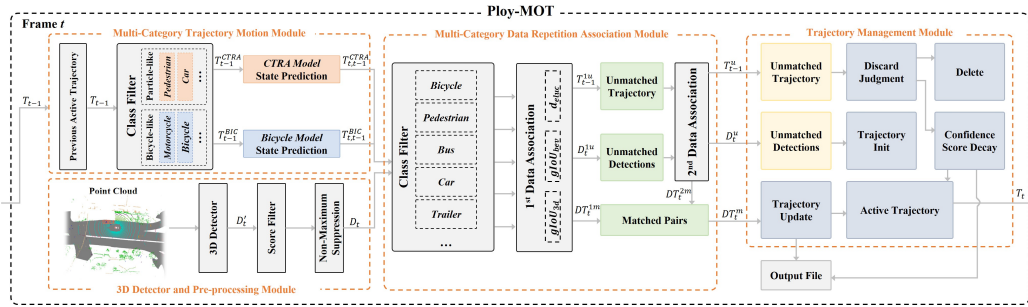


Figure 2: Pipeline of the Poly-MOT framework [54] explaining how the trajectory and object classification works.

### 2.6.1 Pre-processing

Poly-MOT follows a tracking by detection paradigm meaning that the frame first passes through an object detection before object tracking. The detection framework used for this thesis is CenterPoint and includes bounding boxes with very low confident scores. Detection algorithms often deliberately produce many low-confidence score boxes to enhance their recall. However, this is detrimental for the tracking process. Low confidence score increases ID switches, which is when an object loses and regains visibility of an object over a couple of frames and is given another identity. Ideally the object should have the same identity when the object is found again. This is beneficial to get a better understanding of each object's movement. To solve this issue the Poly-MOT has implemented Non-Maximum Suppression (NMS) that removes bounding boxes with similar features (e.g., overlapping bounding boxes or those with really close euclidean distance). NMS require more computational power specifically when using a large dataset like nuScenes

and to reduce overhead Poly-MOT has combined NMS with Score filter (SF) that eliminates every bounding box under a certain score.

By including these two algorithms, the framework has achieved an improvement of 1.3% AMOTA and reduced the run time by 40% compared to using NMS alone. After pre-processing, the center position of the bounding box  $(x, y, z)$ , size  $(w, l, h)$  and heading  $(\theta)$ , and velocity  $(v_x, v_y)$  can be extracted and utilized in the next step [11].

### 2.6.2 Trajectory Motion Models

What distinguishes Poly-MOT compared to other tracking frameworks is its consideration to different object classes and their specific motion characteristics. For instance, pedestrians move differently from trailers and motorcycles. Other frameworks often consider constant velocity or constant acceleration as their only motion model, while Poly-MOT utilizes three different motion models: Constant Turn Rate and Acceleration (CTRA), Bicycle model, and Constant Acceleration (CA). CA motion models are represented by this state vector:

$$\mathbf{T}_{CA} = [x, y, z, w, l, h, v_x, v_y, v_z, a_x, a_y, a_z, \theta]. \quad (24)$$

**The CA model** designed to handle detected trailers utilizes the Constant Acceleration (CA) model which assumes constant acceleration, meaning that the position  $(x, y)$  is updated using the following equations. In (26),  $dt$  represents the time difference, allowing for the calculation of the distance traveled due to the velocity  $v$  and the additional distance due to the acceleration  $a$ . The prior position is represented as  $x_{k-1}$ .

$$x_k = x_{k-1} + v_x \cdot dt + \frac{1}{2} \cdot a \cdot dt^2 \quad (25)$$

$$y_k = y_{k-1} + v_y \cdot dt + \frac{1}{2} \cdot a \cdot dt^2 \quad (26)$$

Since there is no  $v_z$  value, the  $z$  position remains unchanged, as represented by the equation:

$$z_k = z_{k-1} \quad (27)$$

The velocity in x,y can be calculated with:

$$v_k = v_{k-1} + a \cdot dt \quad (28)$$

To use this model for detection of a trailer, the Kalman filter in 3-11 is applied and therefore the hyperparameters  $Q$  (covariance matrix of the process noise) and  $R$  (covariance matrix of the observation noise) needs to be designed. They are set as the identity matrix multiplied with 100 and 0.01, respectively. This signifies a relatively low trusted model, but high trust in the measurement. This is due to the limitations of Linear Kalman filters, it is unable to capture more complex movement due to its relation to linearity and real world examples often deviates from linearity. The CA model then executes the Kalman filter using equations from 3-11 to predict and update the measurement.

**CTRA model:** The Constant Turn Rate and Acceleration (CTRA) model assumes a constant turn rate ( $\omega$ ) and acceleration. The state vector for this motion is represented as follows:

$$\mathbf{T}_{CTRA} = [x, y, z, v, a, \theta, \omega, w, l, h] \quad (29)$$

The heading angle  $\theta$  will have nonlinear dynamics, unlike in the CA model. For this reason the EKF will be used to estimate the state from measurements.

The hyperparameter  $Q$ , which represents the covariance of the process noise, is set to 1 for this motion model. This indicates a higher level of trust in this model compared to the CA model. This higher level of trust is due to the EKF's ability to linearize nonlinear models using the first-order Taylor expansion.

This motion model is used for objects identified as cars, pedestrianized buses, and trucks.

The velocity calculation is given by:

$$v_t = v_{t-1} + a\Delta t \quad (30)$$

And the position calculation is:

$$\hat{x}_{t,t-1} = \hat{x}_{t-1} + \int_{(t-1)\sigma}^{t\sigma} v(\tau) \cos(\eta(\tau)) d\tau \quad (31)$$

## 2 THEORETICAL BACKGROUND

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Where  $\sigma$  is the interval between two frames.

**Bicycle model:** The Bicycle model assumes the steering angle and velocity to be constant but can adjust for the change in heading angle and velocity direction, making it suitable for modeling motorcycles and bicycles. The state vector for the Bicycle model is defined as:

$$\mathbf{T}_{BIC} = [x', y', z, v, a, \theta, \delta, w, l, h], \quad (32)$$

Unlike CTRA that considers the geometric center, the bicycle model calculates around the center of gravity and adjusts for turn rate and steering angle while maintaining a constant velocity:

$$v(\tau) = v_{t-1}$$

The slip angle, which is the angle between the front tire and the velocity in which the object is heading, can be calculated by:

$$\beta(\tau) = \tan^{-1} \left( \frac{l_r}{\gamma l} \tan(\delta(\tau)) \right) \quad (33)$$

where  $\gamma$  is the ratio of the distance between the wheels and the total length of the object,  $l$  is the total length, and  $l_r$  is the length of the center of gravity to the object's rear wheel. Calculating the slip angle is important for better understanding how much the tire deviates from the direction in which the front tire is pointing due to lateral forces.

The angle of velocity with relations to the X-axis can be calculated by:

$$\eta(\tau) = \theta(\tau) + \beta(\tau) \quad (34)$$

This is needed to determine the direction of the vehicle's motion.

The turn rate over time for the Bicycle model can be calculated by:

$$\omega(\tau) = \frac{v(\tau) \sin(\beta(\tau))}{l_r} \quad (35)$$

The heading angle for both CTRA and Bicycle can be calculated by:

$$\theta(\tau) = \theta_{t-1} + \omega(\tau) \Delta t \quad (36)$$

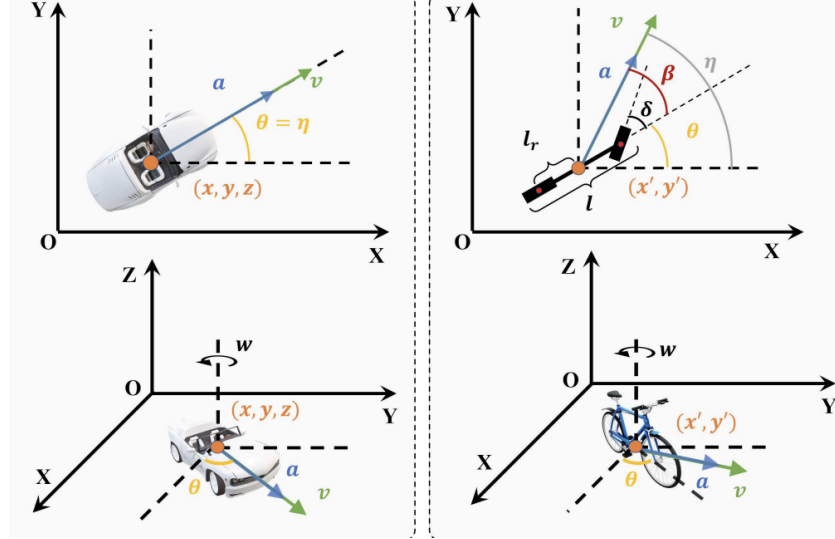


Figure 3: Angle and position representation for Bicycle and CTRA from [11]

### 2.6.3 Data association

Poly-MOT also adjusts the data association depending on different class objects. Due to each class object having a unique geometric shape they become sensitive to different threshold and similarity metrics. This framework produces 3 similarity metrics  $gIoU_{bev}$ ,  $gIoU_{3D}$  and  $d_{eucli}$ . Where  $gIoU_{bev}$  is the ratio of how much a detection bounding box overlaps with trajectory bounding box in bird eye view, 2D space.  $gIoU_{3D}$  is the same but for 3D space and  $d_{eucli}$  is the euclidean distance between trajectory and detection bounding boxes. Together with either the Hungarian algorithm or the Greedy algorithm, Poly-MOT associates detections with the corresponding trajectories by selecting those with the highest  $gIoU$  or the lowest  $d_{eucli}$  depending on the class object. This ensures finding the global optimal similarity metric for a more accurate object tracking

Poly-MOT also has a second association for undetected matches in order to further reduce false positives. For this  $gIoU_{bev}$  is employed to all of the undetected and a strict threshold is applied.

And for all trajectory outputs, a strict threshold of 0.08 is applied to further reduce FN (False Negatives).



### 2.6.4 Trajectory Management

To manage which trajectories to keep and which to be discarded Poly-MOT has created a Trajectory Management that also is unique to each class. The object class has a different "Max age" which is the highest "age" for an object to be detected until it is deleted. There is also a unique decay rate that punishes the trajectory score differently. Bicycle and motorcycle has the highest decay rate due to them being harder to predict, trajectory score under 0.08 will also be deleted.

## 2.7 Autonomous Driving Datasets

There are several different datasets used for developing autonomous vehicle systems, some of the most prominent ones are Waymo, KITTI and nuScenes dataset. These datasets include information of real-world driving environments, sensor data and important sample annotations. Sample annotations include raw data from sensors, bounding boxes information and labels and are used to validate and improve performance.

### 2.7.1 KITTI

The KITTI dataset was made public in 2012 and is one of the early open source datasets which propelled the research in the autonomous field forward [55]. The KITTI dataset includes a 64 beam LiDAR sensor along with 4 cameras with 1384x1032 resolution. Two of the cameras are grayscale and the other two offer color, both sampled at a max frequency of 15 Hz. These sensors also provide information for sample annotations at 1 Hz and provide around a total of 41k sample annotations [55]. KITTI dataset provides rich information on driving environments and makes it well suited for 2D, 3D object detection and object tracking [56]. Having multiple sensors and annotations make it a good choice for evaluation training algorithms.

### 2.7.2 Waymo

Waymo is another open source dataset, including five 64-beam LiDAR sensors. Among these four are short range with a maximum reach of 20 m and one medium range sensor that can reach 75 m. Complementing these five LiDAR sensors are five more color cameras with resolution of 1920x1280.

This dataset comprises 1150 different scene including a wide range of environments such as downtown, suburban, night, rain and construction. Waymo offers 10 hz sampling which provides 230k annotated samples. The substantial amount of annotated samples in different environments make this dataset great for detection and tracking algorithms [57].

### 2.7.3 nuScenes

nuScenes is a public large-scale dataset for autonomous driving [58]. The data provides recordings of the streets of Boston and Singapore, providing data in different traffic environments, right and left-hand traffic. nuScenes is one of the few datasets that provides 3 sensor modalities containing, 6 cameras, 5 Long range radars, and 1 mechanical LiDAR providing a 360° field of view. The full dataset includes 1.4M camera images, 390k LiDAR sweeps, 1.4M radar sweeps, 1.4M object bounding boxes, and 40k sample annotations.

The various modalities contain different range intervals. LiDAR, which is a spinning sensor consisting of 32 beams with a capture frequency of 20 Hz has a max range of 70m. The Frequency modulated continuous wave radar sensor has a 13Hz capture frequency and a max 250m range. The Camera is a complementary metaloxide semiconductor-sensor, RGB-camera of 12Hz frequency and 1600x900 resolution [58].

Each scene of the data consists of 20s duration each. Such scenes contain various scenarios which include high traffic density, dangerous traffic situations as well as 19% different weather and lighting conditions, and 12 % during the night time [58].

The thesis has decided to use nuScenes as the dataset even though Waymo offers a lot more annotated data. The reason is because it includes radar sensor data and the goal of the thesis is to explore multi sensor integration with sensor fusion to evaluate if the radar sensor can enhance object tracking.

### 2.7.4 Evaluation Metrics

nuScenes evaluates and compares the results with different evaluation metrics depending on the specific task. This dataset supports multiple tasks

including detection, tracking, prediction, and localization. There are various metrics that nuScenes uses, for object tracking there are:

**False positives (FP)** and is when the tracking algorithm identifies a non-existing trajectory in the current frame. Common examples of FP are multiple bounding boxes for the same object or identifies a shadow as a pedestrian.

**False negatives (FN)** is when the tracking algorithm is unable to find a track a detected object. Common examples of FN are when the tracking algorithm is unable to match trajectories with the right object.

**True Positive metrics(TP)** is when the model is able to correctly track the detected objects [58].

**Identity switches (IDS)** a metric indicating how many times a tracked object's identity has switched. This metric is valuable when used analyzing the accuracy and consistency of the object tracking algorithm. A high number of IDS will indicate a a object tracking model that has difficulties to correctly maintains the same tracking ID over time and subsequently a high IDS can also indicate an inaccurate model, as the object can be associated with the wrong identity.

**Recall** is a metric that is commonly used in object tracking; it helps to quantify the effectiveness of the object tracking algorithm's ability to track and identify objects correctly.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (37)$$

**Average Multi Object Tracking Accuracy(AMOTA)**. The classical metrics methods do not take into account the confidence of a prediction, leading to the development of AMOTA. AMOTA takes the number of ground-truth positives for a current class and evaluates how well the system is able to track multiple objects over time. AMOTA is a metric that considers false positives, true positives, false negatives and identity switches.

Equation for *AMOTA* with Recall:

$$AMOTA = \frac{1}{l} \sum_{r \in \{\frac{1}{l}, \frac{2}{l}, \dots, 1\}} \left( \max \left( 0, 1 - \frac{FP_r + (1 - \text{Recall}) \cdot P}{\text{Recall} \cdot P} \right) \right) \quad (38)$$

P represents the precision and is defined as the ratio of the true positives and the total number of positive detection. While the parameter  $r$  reflects the importance of precision. This is often set as a high value to punish low precision, but when the evaluation is more focused toward lowering IDS and FN a lower  $r$  can be chosen to balance the importance of precision and IDS and FN.

Here,  $l$  is the number of recall values considered, and the sum is taken over  $l$  values of  $r$ .

## 2.8 Rigid Body Transformation

During this thesis various coordinate frames were used to analyze and compare data. To transition between these frames, rigid body transformations were necessary. For example, when aligning sensor data from different sources, rigid body transformations allowed us to place all data within a common reference frame for accurate comparison and analysis.

Rigid transformations exhibit two essential characteristics: maintaining absolute distances between points and preserving orientation and area (or volume in 3D). The transformation equation involves a rotation matrix ( $\mathbf{R}$ ), an arbitrary translation ( $\mathbf{t}$ ), and the coordinate system to be transformed ( $\mathbf{x}$ ) [59, 60]. The Rigid body transformation given by

$$\mathbf{p}_{trans} = \mathbf{T}\mathbf{p}$$

introduces  $p$  and  $p_{trans}$ , which are the original and transformed points in homogeneous coordinates, respectively, and  $T$  the transformation matrix. Diving deeply into the transformation matrix, we can observe that it is composed by two key variables the Rotation matrix  $R$  and the translation matrix  $t$ :

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Which can be simplified as:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \mathbf{R} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \mathbf{t}$$

Rigid transformations allows transitions between various coordinate frames. Consider the following example with the coordinate frames  $A$ ,  $B$ , and  $C$ :

$$\mathbf{T}_{AB} = \begin{bmatrix} \mathbf{R}_{AB} & \mathbf{t}_{AB} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad \mathbf{T}_{BC} = \begin{bmatrix} \mathbf{R}_{BC} & \mathbf{t}_{BC} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

Where  $T_{AB}$  is the transformation matrix from frame  $B$  to frame  $A$ , and  $T_{BC}$  is the transformation matrix from frame  $C$  to frame  $B$ . By combining these transformations, a combined transformation from frame  $C$  to frame  $A$  is achieved:

$$\mathbf{T}_{AC} = \mathbf{T}_{AB} \mathbf{T}_{BC}$$

$$\mathbf{T}_{AC} = \begin{bmatrix} \mathbf{R}_{AB} & \mathbf{t}_{AB} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{BC} & \mathbf{t}_{BC} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

Thus, the resulting rotation matrix and translation vector are:

$$\mathbf{R}_{AC} = \mathbf{R}_{AB} \mathbf{R}_{BC}$$

$$\mathbf{t}_{AC} = \mathbf{R}_{AB} \mathbf{t}_{BC} + \mathbf{t}_{AB}$$

This demonstrates the practicality of rigid transformations for moving between coordinate frames, a technique extensively utilized in this thesis to ensure all measurements were comparable and unified in the same coordinate system.

## 2.9 Data Aggregation

Aggregation of data refers to the method where raw data is gathered and used in a summary form for data analysis. This method often involves gathering data over a defined period to extract additional insights about specific objects or characteristics. A key aspect of this process is Time Aggregation, where data points are gathered over a specified time period [61].

This process is particularly useful for enhancing the density of sensor data points. In the context of the nuScenes dataset, where the radar points are known to be sparse, temporal data aggregation helps address this limitation. By incorporating information from multiple time steps, this technique fills the gaps, reduces the measurement noise and contributes to a more comprehensive understanding of the objects surrounding the vehicle.

However, data aggregation has issues. Firstly, the process to increase the data density may unintentionally lead to data redundancy, where identical points are stored repeatedly. Secondly, the increase of data points leads to computation overhead, requiring more time and memory to perform certain tasks [62].

## 3 Related Work

This section provides useful knowledge and an overview of the current State Of The Art (SOTA) object detection and tracking models in ADAS.

Jahromi et al.'s work [13] demonstrates a hybrid fusion framework integrating LiDAR/camera sensors, employing qualities of deep learning algorithms and Extended Kalman Filter (EKF) techniques to increase the model's accuracy. The approach consists of two main parts. Firstly, the model fuses the raw video frames from the camera sensor along the depth information from the LiDAR using Deep Neural Network (DNN). This initial stage manages the object classification and road segmentation task. CNN and FCN are often used to tackle these challenges, demonstrating superior performance

### 3 RELATED WORK

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in computer vision tasks compared to traditional methods. During the second stage, LiDAR- and radar-data are used to tackle object-tracking tasks. EKF plays a key role by fusing PCD and radar signals at high-level fusion. Given the vehicle’s non-linear movement, the EKF solves the challenge beyond the limits of traditional Kalman filter, hence obtaining highly accurate object-tracking.

The work of Xuanyao et al. [63] uses the Futr3d framework, which is an unified end-to-end sensor fusion framework for 3D detection. This sensor pipeline integrates data from multiple modalities, including cameras, LiDAR, and radar available on the nuScenes database. The research introduces a query-based Modality-Agnostic feature sampler (MAFS) framework that individually extracts features from each modality and employs a transformer decoder for comprehensive processing, as shown in Figure 4. The results demonstrate a flexible model yielding results comparable to SOTA models e.g. CenterPoint, achieving a 58.8 mean Average Precision (mAP).

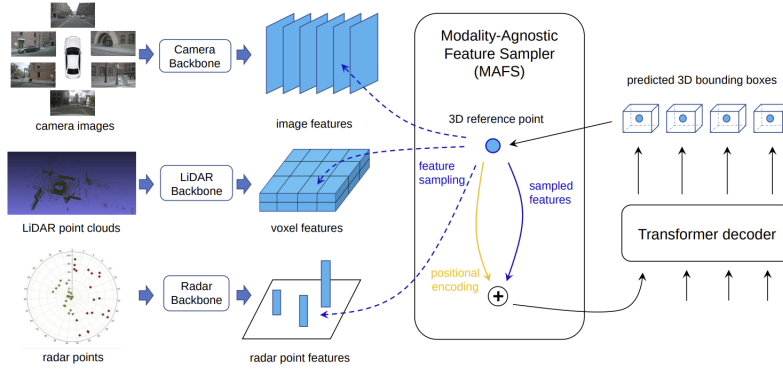


Figure 4: Futr3ds model approach implemented in [63]

The research of Tianwei Yin et al. [11] introduces Centerpoint, a detection framework built on LiDAR-based backbone networks Voxelnet [38] and PointPillars [64]. Centerpoint detects the center of objects and extracts key attributes such as 3D size, orientation, and velocity later used to improve the detection performance. This framework simplifies objects as points, making it possible to predict features like velocity between consecutive frames, which are then used to facilitate the tracking task. As illustrated in Figure 5, Centerpoint uses a 3D backbone to extract map-view features from LiDAR

PCD. Subsequently, a 2D CNN architecture to find the center of objects is employed. followed by Multi-Layer Perceptron (MLP)-evaluation to assign a confidence score for each detection. Centerpoint’s performance is tested on nuScenes Dataset, showcasing SOTA results of 65.5% nuScenes Detection Score (NDS) and 58.0% mAP

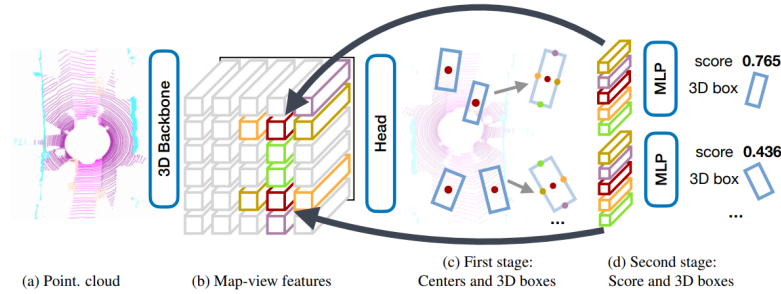


Figure 5: Overview of CenterPoint framework [11]

Aleksandr Kim et al. work EagerMOT is a framework that integrates sensor fusion with camera and LiDAR detection [65]. This framework fuses 3D detection from LiDAR with 2D detections from camera, merging the result from the same object. This high level sensor fusion is beneficial when one sensor fails; for example when the LiDAR sensor is affected by occlusion and is unable to detect the object, the camera might still be able to detect the object thus improving robustness. EagerMOT utilizes a single motion model, the constant velocity combined with a Kalman filter to predict the trajectory’s future position of the detected object. EagerMOT includes two stage association, first stage uses 3D Intersection over union (IoU), while second stage data association addresses the unmatched trajectories with 2D IoU. Both association stages utilize the Hungarian algorithm to get the global value (highest value of overlap), with strict threshold ( $\theta$ ) to filter out really bad matches. The life cycle management of EagerMOT will discard trajectories that are not updated in a specified number of frames. This framework has incorporated the PointGNN 3D detector and TrackRCNN 2D detector, achieving 0.68 AMOTA on nuScenes dataset.

Simple track developed by Ziqi Pang et al. [42], is another SOTA object tracking model, able to achieve 0.696 AMOTA. This framework utilizes non-maximum suppression (NMS) to filter out the low quality bounding boxes.



Simple track utilizes the Kalman filter to predict the future position of the detected objects. It also compares the results using Hungarian and Greedy algorithms with different similarity metrics such as IoU and Generalized intersection over union (GIoU), but ultimately deciding to use GIoU and Hungarian algorithm because it performed slightly better.

Simple Track uses two stage association, with two strict thresholds of 0.5 and 0.1 respectively, only associating scores higher than these. Importantly, the second association will not be included in the final output but is instead kept alive and continually predicts the position until the score becomes high enough (0.5) or falls under 0.1 and is then deleted. This is to reduce FNs and decrease IDS without affecting AMOTA.

## 4 Methodologies and Methods

The object tracking framework Poly-MOT [54] described in Section 2.6, specializes in object tracking using LiDAR data. During this thesis, we will explore the capabilities of the radar sensors by creating a sensor fusion method between LiDAR and radar data using the bayesian method Kalman Filter. We will investigate the impact of this fusion on the tracking performance of the framework.

### 4.1 Radar Data Integration and Transformation

The data used in this thesis is composed of five radar sensors strategically placed around the vehicle to provide a 360° view of its surroundings. Each sensor is uniquely labeled as radar-front, radar-left, radar-right, radar-back-left, and radar-back-right, allowing an extensive analysis of objects from different point of views. The radar sensors detect the position and velocity of objects, which are crucial for understanding the movement and tracking of these objects. The annotations accompanying the sensor data enhances the understanding of each data sample, providing information of objects size, rotation, and scene descriptions, including environmental conditions such as rain, night, or clear weather. Another important and useful information is the *ego-pose* which contains information about the vehicle’s position and orientation relative to its environment.

As described in Section 2.4.1, the Kalman filter requires measured data to

## 4 METHODOLOGIES AND METHODS

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initialize and update the filter for predicting object positions. In this thesis, radar data is used to obtain object velocity from the Doppler effect, while LiDAR sensor data provides information on the position, size, rotation, and velocity of each object. To accurately associate these data sources, we will use the Hungarian Algorithm, as described in Section 2.5.1, to match radar data points with LiDAR detection data effectively.

The nuScenes dataset provides three distinct coordinate systems: sensor-pose, ego-pose and global-pose. Where each contains translation and rotation information for each sample. For instance, sensor-pose provides information about the sensor placement relative to the vehicle, as displayed in Figure 6 where each sensor has its unique coordinate system.

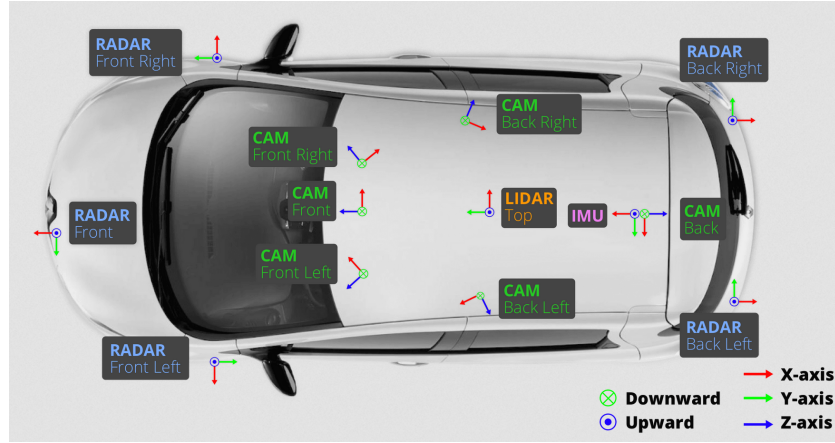


Figure 6: Sensor placement and coordinate orientation on the nuScenes vehicle, visualized in [58].

The ego-pose displays the position of the vehicles rear axle projected onto the flat ground [59]. The global-pose contains the map coordinates in meters, origo being the top left corner of the map provided by nuScenes. To transition between the various coordinate frames given by nuScenes, the rigid body transformation described in Section 2.8 is applied. The raw data from the nuScenes dataset is given in the sensor-pose coordinates, whereas our framework operates in the global-pose coordinate system. By using the rigid body transformation, it is possible to transform the data to the desired coordinate system. This transformation is performed by using various transformation matrix  $T$  that allows the transition between coordinate frames given by

$$\mathbf{T}_{sensor}^{global} = \mathbf{T}_{ego}^{global} \mathbf{T}_{sensor}^{ego},$$

where the global coordinate system is achieved by transforming data from the sensor coordinate system to the ego coordinate system, and then to the global coordinate system.

One particular challenge that nuScenes dataset faces is the sparsity of radar data. In response, this work introduces the implementation of data aggregation, particularly temporal data aggregation as described in Section 2.9. This approach enables the storage of radar data from previous time frames, creating a historical dataset containing radar measurements to address the gaps and enhance the density of the radar points data.

One method to enhance the sparse radar points is to use aggregated data. The polling period, meaning the time interval for data gathering from sensors, is normally 0.5 s. By implementing data aggregation, the number of radar points within bounding boxes increases significantly. Although this amplifies computational overhead, its application leads to improving the data information obtained from radar sensors. This improvement is illustrated in the comparison between the Figures 8 and 9, which shows pre- and post-data aggregation, highlighting an increase in the available data for further analysis.

The integration of data aggregation into the framework is outlined in Algorithm 1, where  $n_{samples}$  is defined as the amount of samples used for data aggregation. A time difference is calculated between the current data sample and the previous one, adding a temporal feature to the data. This feature allows the model to understand the temporal context of each data point, enabling the model to differentiate points from different time steps to help understand the movement and dynamics of objects over time.

This approach also helps improve the tracking of objects by incorporating time-based information. Once the data is combined into a single matrix, it is then used for data association, where each point is matched with a corresponding object bounding-box. For all the points inside a box, the average value of the point information is calculated, achieving a comprehensive information about each object.

**Algorithm 1** Data Aggregation and Data Association

---

**Input:**  $n_{\text{samples}} \leftarrow$  Number of samples**Output:** Aggregated Radar data with temporal features**for**  $i$  to  $n_{\text{samples}}$  **do**     $current\_pc \leftarrow$  Current sample radar data     $time\_lag \leftarrow$  time difference between the current and previous sample    Aggregate the  $current\_pc$  points with the previous data to form  $aggregated\_data$ **end for**

Perform Data Association

**for** each Bounding-Box  $b$  **do**     $points\_in\_box \leftarrow []$   $\triangleright$  Initialize empty list for points in the box    **for** each point  $p$  in  $aggregated\_data$  **do**        **if**  $p$  is inside box  $b$  **then**             $points\_in\_box \leftarrow p$         **end if**    **end for**     $mean\_value \leftarrow$  Calculate the mean of  $points\_in\_box$  for each object**end for****Return** Mean value for all the aggregated data points for each box

---

## 4.2 Radar Sensor Fusion

To effectively use the extracted radar data, each radar point must be assigned to a specific bounding box. Our method ensures that only radar points within a bounding box contribute their values to that box. For radar association, we have implemented the following two approaches:

The first approach uses Hungarian algorithm to solve a cost matrix constructed with euclidean distance for each radar point and detection bounding box. Then select the radar point that is nearest to the bounding box center position. Only the radar point that lies within the bounding box are selected when calculating for the euclidean distance, see Algorithm 2.

The second approach is more suitable when there are a lot of radar points inside a bounding box and returns the average value of all the radar position and velocity. This method is effective in mitigating the effects of noise com-

monly associated with radar sensors and can be useful when using aggregated data, see Algorithm 1.

Upon the completion of either the first or second approach, the radar value is integrated to the data and is forwarded to the Kalman filter for further processing.

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**Algorithm 2** Object-Radar Matching using Hungarian Filter

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**Require:**

*Detection\_Data*: Data containing object information (size and pos)

*Radar\_Data*: Data containing radar positions

*det\_num*  $\leftarrow$  Number of objects

*rad\_num*  $\leftarrow$  Number of radar points

**Ensure:** Updated object data with matched radar positions

*cost\_matrix*  $\leftarrow$  array of size  $[det\_num, rad\_num]$

**for**  $i = 1$  to *det\_num* **do**

    Extract object information (size and position).

**for**  $j = 1$  to *rad\_num* **do**

        Extract radar position.

**if** Radar position falls inside object's area **then**

            Euclidean distance between radar point and object's center

*cost\_matrix* $[i, j] \leftarrow$  Euclidean distance

**else**

            Set *cost\_matrix* $[i, j] \leftarrow \infty$

**end if**

**end for**

**end for**

Use the Hungarian algorithm to find optimal match of radar points to objects based on *cost\_matrix*.

**for all**  $(i, j)$  in matched pairs **do**

    Update *object\_data* $[i, 14 : 17]$  with *radar\_pos* $[j]$ .

**end for**

---

### 4.3 Kalman Filter and Motion Model

In this master thesis we have further developed the Kalman filter and motion model module of Poly-MOT to integrate radar data.

This thesis approach is to fuse the velocity values based on both the radar and the LiDAR measurement in order to use both information. This approach utilizes both sensors by adapting the hyperparameters of the Kalman filter to account for the reliability of each sensor.

In the single measurement update approach, the measurements from the multiple sensors are incorporated simultaneously. For combining the measurement into a single update step the vector is constructed using the observation model  $H_k$  as

$$z_k = H_k x_k + v_k, \quad (39)$$

where the measurements of the LiDAR sensor and radar sensors are combined in matrix with each sensor's data remaining independent of the other represented by

$$\mathbf{z}_k = \begin{bmatrix} \mathbf{z}_{\text{LiDAR}} \\ \mathbf{z}_{\text{Radar}} \end{bmatrix} \quad \mathbf{H}_k = \begin{bmatrix} \mathbf{H}_{\text{LiDAR}} \\ \mathbf{H}_{\text{Radar}} \end{bmatrix},$$

thus leading to a noise covariance matrix of  $v_k$  as

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{\text{LiDAR}} & 0 \\ 0 & \mathbf{R}_{\text{Radar}} \end{bmatrix},$$

the Kalman filter will then perform the update step as described in section 2.4.1.

#### 4.4 Bad Weather Split

The nuScenes dataset provides various splits for training (train-split), validation (val-split), and mini splits for an initial understanding of the data (mini-train, mini-val). During this thesis, a weather split was created to further investigate the impact rain has on the results. This split includes data scenes recorded during rainy weather conditions to analyze the performance of both radar and LiDAR sensors during these conditions.

The *weather-split* is composed of 27 scenes for validation, all portraying moderate rain conditions. In comparison, the standard nuScenes validation split comprises 158 scenes, meaning the weather split represents less than 20% of the total validation data.

To evaluate the results for this split, a separate evaluation was created to only take into account the rainy-scenes. This approach enables a targeted analysis of the framework’s performance under specific weather conditions.

## 5 Benchmark and Evaluation

This section will present the experimental setup and the evaluation metrics used with the Poly-MOT framework serving as the benchmark to surpass. We also emphasize the importance of having a good velocity estimate and to highlight the amount of usable radar points in nuScenes dataset to enhance the object tracking information.

### 5.1 With and Without Velocity

To benchmark the results of this thesis, the Poly-MOT framework was initially evaluated, both with and without the velocity, with the default velocity being derived from LiDAR. This will in later sections be compared with the integrated radar sensor data through sensor fusion. This thesis will explore how the integration of radar sensor velocity data affects the performance of the framework.

As illustrated in the table 2, it is clear that having a velocity measurement does improve the AMOTA for every class object and further minimizes IDS and FN while also increasing the TP results.

Figure 7 further illustrates the impact of having velocity data. Velocity data improves the model’s prediction, results of the incorporated velocity data predicts a much closer bounding box to the ID 74 car than the counterpart, non-velocity data, results. For a car object the  $GiOU_{3d}$  threshold is 1.31 meaning that the trajectory and the detection need to overlap with at least 31%. Without velocity data the model was still able to find a match but barely. A slightly worse prediction would have resulted in lost trajectory.

| AMOTA |         |      |      |      |      |      |      |      |      |       |       |       |         |
|-------|---------|------|------|------|------|------|------|------|------|-------|-------|-------|---------|
| Velo  | Overall | Bic  | Bus  | Car  | Moto | Ped  | Tra  | Tru  | IDS↓ | FP↓   | TP↓   | FN↓   | Recall↓ |
| YES   | 73.0    | 57.1 | 87.9 | 85.7 | 76.4 | 81.4 | 50.7 | 71.4 | 302  | 14040 | 84358 | 17237 | 75.8    |
| NO    | 70.7    | 56.9 | 85.2 | 84.4 | 68.7 | 81.3 | 49.2 | 69.5 | 418  | 13472 | 83295 | 18184 | 73.6    |

Table 2: Results of the Poly-MOT Benchmark comparing scenarios with and without the inclusion of velocity in the calculations.



## 5 BENCHMARK AND EVALUATION



Figure 7: Camera visualization of the benchmark results including velocity (top) and without including velocity (bottom) in the calculations.

### 5.1.1 Radar Point Distribution

As shown in Figure 8 the number of radar points found inside a bounding box per sample is presented. This data shows that most samples contain 0-19 radar points that can be found inside a bounding box and can provide additional information. An analysis will determine if these radar data can improve the robustness and accuracy of object tracking.

Figure 9 illustrates how the implementation of data aggregation affects the number of radar data points within a bounding box for 2 and 4 samples respectively. A "sample" in this context refers to a single instance of synchronized sensor data collected at a specific time point. Each radar point inside a sample contains information about a specific objects position and velocity detected by the radar sensor. The top figure shows the majority of points within the range of 0-25 points/sample, decreasing towards 70 points/sample. In the case of 4 sample data aggregation, it can be observed a general increase of data points across all ranges compared to the previous figures.

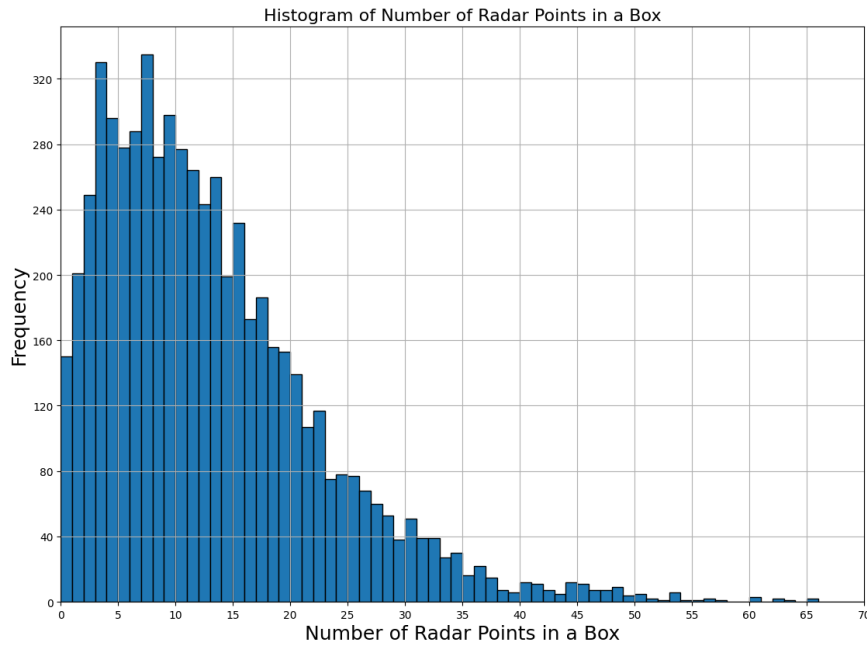


Figure 8: Number of radar points in a bounding box per sample

## 5 BENCHMARK AND EVALUATION

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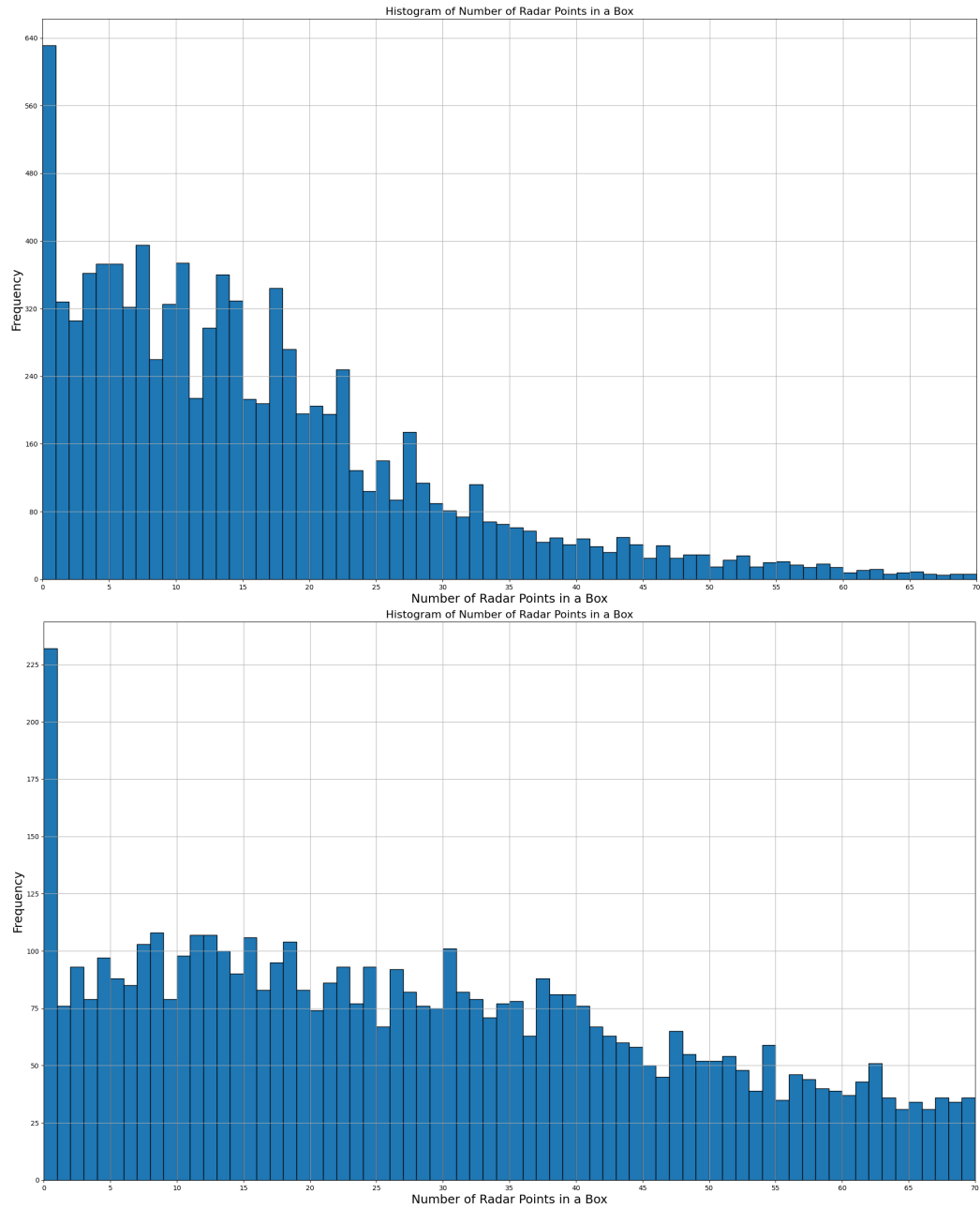


Figure 9: Number of radar points in a bounding box per sample, while aggregating data for 2 samples (top) and aggregating data for 4 samples (bottom)

## 6 Experimental Results

This section will analyze the findings of this thesis by examining the evaluation metric from object tracking and the predicted motion for an object in Bird-Eye View (BEV). It will also include figures of the tracking results from the camera point of view, to further analyze the performance of the object tracking algorithm.

Furthermore the sensor fusion with LiDAR and radar will be evaluated on both the *validation-split* and *weather-split*. This evaluation will determine if the radar sensor can improve object tracking in normal and challenging weather conditions. Figure 10 illustrates this thesis addition to the PolyMOT framework.

## 6 EXPERIMENTAL RESULTS

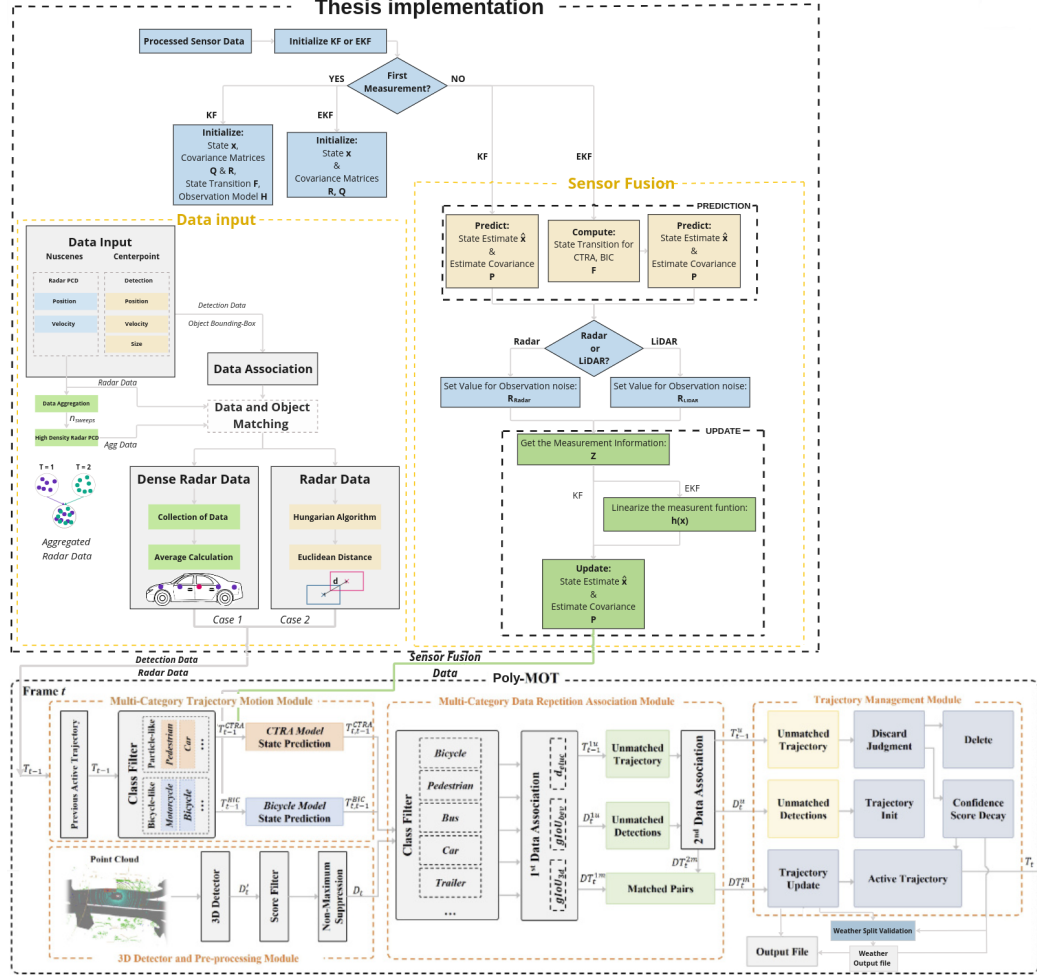


Figure 10: Overview of the works implementation in the Poly-MOT framework, incorporating radar data with additional features such as data aggregation and Euclidean Distance for data filtering. The Sensor Fusion method employs two types of filters, the *Kalman Filter* and the *Extended Kalman Filter*.

### 6.1 Radar Sensor Fusion Results

The Kalman filter's performance of the framework is illustrated in Figure 11. The image demonstrates the ability of the filter to update its information based on the position and velocity detection information and even predict

future states when the detected object no longer is available.

As highlighted in Table 3, integrating radar sensor velocity data did not improve the object tracking results on the validation dataset. By adjusting the observation covariance noise,  $R_{Radar}$ , it was observed that higher values (thereby minimizing the impact of radar sensor values) improved the results compared when  $R_{Radar}$  was set to 0.1. When the Kalman filter trust the radar data more the IDS and FP increased and the Recall score was decreased. In contrast, when  $R_{Radar}$  was set to 10 the radar sensor values becomes less relied upon and therefore having a less of an impact, resulting in a better result for the overall performance.

Furthermore Table 3, reveals that using only the radar sensor for the velocity measurement affects negatively the AMOTA performance, similar to the scenario where no velocity measurement was included at all. This result shows that the radar sensor can not perform as the stand alone sensor and needs to be complemented with LiDAR sensor. The radar sensor alone is noisy and has issues measuring velocity in certain angles from radar sensor to the velocity and will be further analyzed in Section 7.

On the other hand, using aggregated data demonstrates clear improvements in the performance on the radar dataset. Although the AMOTA does not improve, it is able to match the performance of the Lidar-based Poly-MOT framework and even improve other evaluation categories. Specifically, Table 3 shows improved results for TP (84548 compared to 84358) and FN (17219 compared to 17237). When comparing aggregated data to non-aggregated data results, improvements are evident across all parameters. It is observed an overall improvement of 0.5% in AMOTA, significant decrease in ID switches from 427 to 304, and an overall enhancement in model performance. This suggests that implementing aggregated data in the nuScenes radar dataset improves model performance.

Additionally, figure 12 illustrates a good example of a predicted movement for a car with and without radar values. When  $R_{Radar}$  was set to 0.1, the predicted trajectory and the car's position was a slightly worse compared to when  $R_{Radar}$  is set to 10. With  $R_{Radar}$  set to 10 being more accurate. But when comparing 12 (top) and 13 they are almost the same, indicating that the aggregated radar data provides more accurate information.

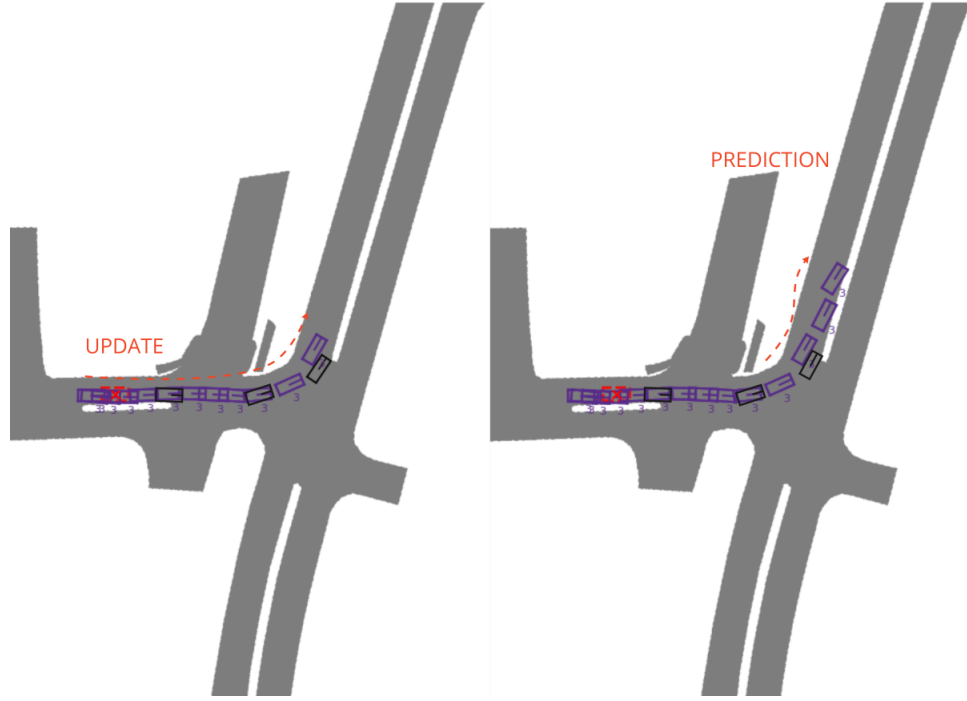


Figure 11: BEV plot of the update and predicted steps of the Kalman filter

| Variables |         |       | AMOTA   |      |      |      |      |      |      |      |      |       |       |       |         |
|-----------|---------|-------|---------|------|------|------|------|------|------|------|------|-------|-------|-------|---------|
| Agg       | $R_L$   | $R_R$ | Overall | Bic  | Bus  | Car  | Moto | Ped  | Tra  | Tru  | IDS↓ | FP↓   | TP↓   | FN↓   | Recall↓ |
| 0         | 1       | 0.1   | 71.0    | 56.5 | 83.7 | 83.1 | 72.8 | 81.3 | 50.7 | 68.9 | 910  | 16433 | 84548 | 16433 | 76.5    |
| 0         | 1       | 1     | 72.4    | 56.7 | 87.5 | 85.0 | 74.7 | 81.2 | 50.7 | 71.0 | 427  | 14488 | 84234 | 17236 | 75.0    |
| 0         | 1       | 10    | 72.8    | 57.1 | 87.7 | 85.7 | 75.7 | 81.2 | 50.7 | 71.3 | 313  | 14159 | 84365 | 17219 | 75.9    |
| 0         | 1000000 | 1     | 70.4    | 56.0 | 81.6 | 83.1 | 72.1 | 81.4 | 50.4 | 68.3 | 933  | 16074 | 84005 | 16959 | 75.4    |
| 2         | 1       | 1     | 72.9    | 57.1 | 87.7 | 85.7 | 76.3 | 81.4 | 50.7 | 71.1 | 304  | 14154 | 84374 | 17219 | 75.8    |
| 4         | 1       | 3     | 72.8    | 57.1 | 87.7 | 85.7 | 76.3 | 81.4 | 50.7 | 70.9 | 310  | 14133 | 84367 | 17220 | 75.8    |

Table 3: AMOTA results for the sensor fusion framework with varying noise covariance  $R$  values, illustrating the impact of different data on performance metrics under normal weather conditions.

## 6 EXPERIMENTAL RESULTS

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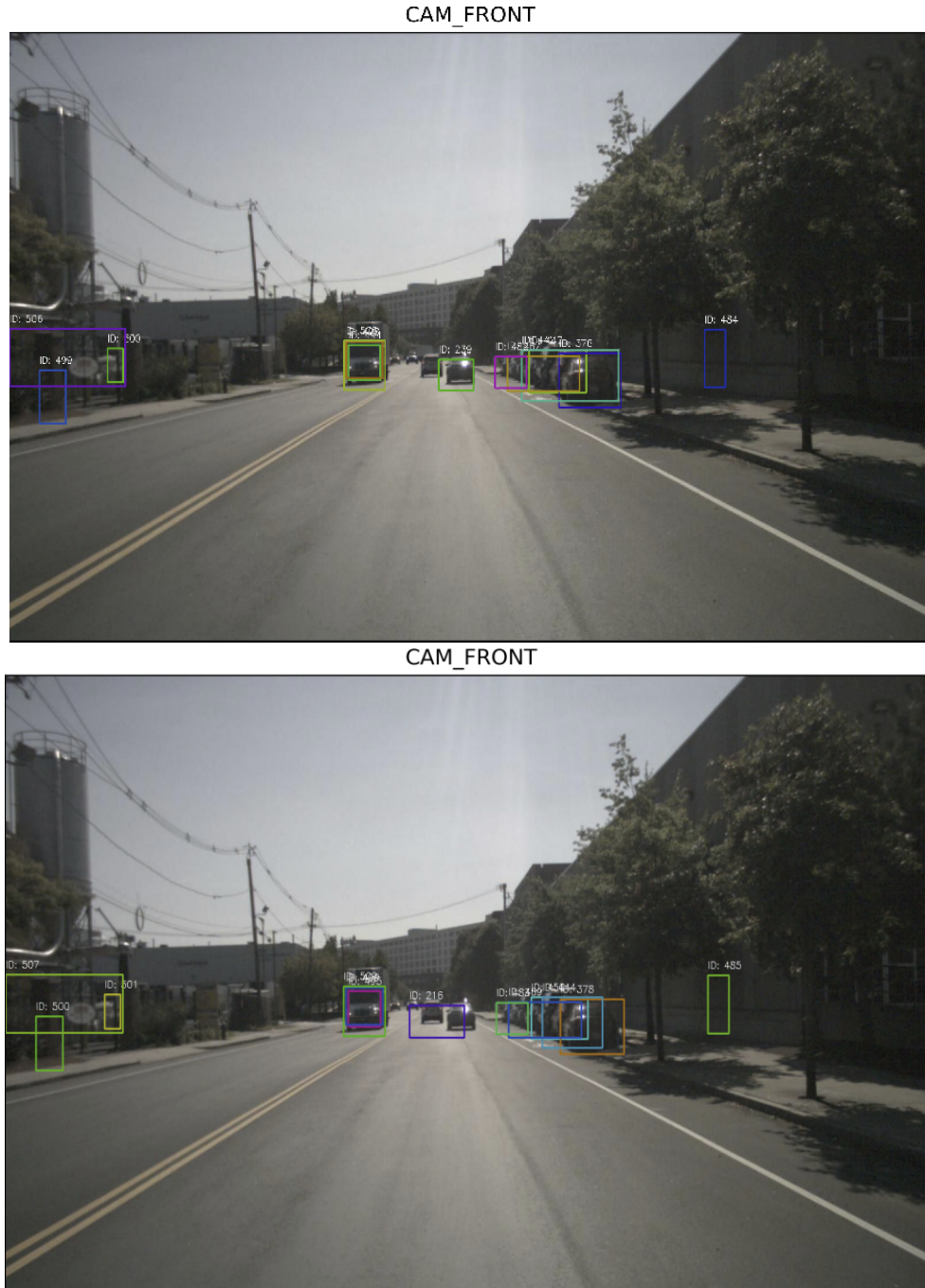


Figure 12: Camera view of sensor fusion results with noise covariance  $R_{Radar}$  set to 10 (top) and 1 (bottom), illustrating object detection and ID classification.





Figure 13: Camera view for sensor fusion results using aggregated data with Noise Covariance  $R_{Radar}$  set to 1, illustrating object detection and ID classification.

### 6.1.1 Bad Weather Split

For the bad weather split, the scenes included are environments with moderate rain and does not include heavy downpours. The results indicate that the rain clearly affected the performance of the LiDAR sensor, almost performing equally bad compared to the radar sensor in these conditions.

Additionally, an important observation from the table 4 ,  $R_{Radar} = 0.1$  (implying higher trust in radar sensor velocity values), shows that the AMOTA for pedestrians increases. This suggests that radar sensor values can help to enhance object tracking’s performance in rainy weather conditions for smaller pedestrian like objects.

When the velocity measurement was solely derived from the radar sensor the AMOTA result was worse implying. This suggests that the LiDAR sensor was still able to perform better than the radar sensor even in rainy condi-

tions.

Analyzing the results from aggregated data reveals that for  $Agg = 2$  the overall AMOTA performs as well as the LiDAR-based framework, achieving a score of 68.9%. Additionally, a comparison between non-aggregated data and aggregated data results shows an improvement in the overall tracking performance.

| Variables |         |       | AMOTA   |      |      |      |      |      |      |      |      |      |       |      |         |
|-----------|---------|-------|---------|------|------|------|------|------|------|------|------|------|-------|------|---------|
| Agg       | $R_L$   | $R_R$ | Overall | Bic  | Bus  | Car  | Moto | Ped  | Tra  | Tru  | IDS↓ | FP↓  | TP↓   | FN↓  | Recall↓ |
| 0         | 1       | None  | 68.9    | 27.6 | 94.5 | 87.9 | 64.9 | 79.3 | 54.6 | 73.6 | 21   | 2102 | 13987 | 2581 | 77.8    |
| 0         | 1       | 1     | 68.4    | 27.6 | 92.1 | 87.2 | 64.9 | 79.5 | 54.6 | 73.2 | 36   | 2202 | 13975 | 2578 | 77.7    |
| 0         | 1       | 10    | 68.6    | 27.6 | 92.1 | 87.9 | 64.9 | 79.3 | 54.6 | 73.6 | 24   | 2116 | 13980 | 2585 | 77.7    |
| 0         | 1       | 0.1   | 68.0    | 27.6 | 92.0 | 85.9 | 63.0 | 80.0 | 54.6 | 72.9 | 96   | 2364 | 14005 | 2488 | 76.2    |
| 0         | 1000000 | 1     | 66.2    | 27.6 | 89.3 | 85.7 | 56.6 | 79.6 | 52.4 | 71.9 | 100  | 2164 | 13678 | 2811 | 75.8    |
| 2         | 1       | 1     | 68.9    | 27.6 | 94.5 | 87.9 | 64.9 | 78.9 | 54.6 | 73.6 | 23   | 2121 | 13977 | 2589 | 77.8    |
| 4         | 1       | 3     | 68.8    | 27.6 | 94.5 | 87.9 | 64.9 | 78.9 | 54.6 | 73.3 | 25   | 2375 | 14239 | 2325 | 78.1    |

Table 4: AMOTA results for the sensor fusion framework with varying noise covariance  $R$  values, illustrating the impact of different data on performance metrics under rainy weather conditions.

## 7 Discussion

To accurately model the real-life dynamics of the object being tracked, optimal filter tuning is essential. This thesis focuses on determining the optimal hyperparameters by testing different values of  $R_k$ . Through systematic experimentation, the value of  $R_k$  yields the best performance has been identified and will be discussed in Section 7.1.

### 7.1 Discussion of Results

By analyzing Table 3 and when the hyperparameter is  $R_r = 0.1$ , the AMOTA decreases with 2 points compared to LiDAR only based method, despite the

increase in Recall, true positives and decrease in false negatives. This is primarily due to the large number of IDS and the increase in false positives. The results indicate that the velocity measurement from radar is highly inconsistent which negatively affects the motion prediction.

When the hyperparameter is  $R_r = 1$  the AMOTA is decreased by 0.6 points, compared to LiDAR only based method. This indicates that the radar velocity measurement is not as trustworthy as the LiDAR velocity measurement and negatively affects the Kalman filters ability to estimate the velocity thus predicting a worse position.

Analyzing when the hyperparameter is instead set to  $R_r = 10$  the AMOTA only decreases by 0.2 points, compared to LiDAR only based method and is because the Kalman filter in most cases only uses the LiDAR measurement and disregards the radar measurement.

When the hyperparameter is set to  $R_r = 1$  and  $R_l = 1000000$  the LiDAR velocity measurement is disregarded and the AMOTA decreases with 2.6 points, compared to LiDAR only based method further proving that the radar velocity measurement is not as reliable.

To address the issue of sparse radar data in the nuScenes dataset, data aggregation was implemented. As shown in Table 3, this approach clearly improves the performance of the radar fusion. Although the improvement does not surpass the performance of the LiDAR-based Poly-MOT framework, it narrows the performance gap considerably.

Data aggregation demonstrated improvement on the performance results compared to non-aggregated data. This improvement is largely due to the method by which aggregated data is integrated into the framework. As demonstrated in Figure 10, aggregated data is used to get a better representative velocity estimate for the detected object, the aggregated approach uses the average data point of all radar points within each object bounding box. On average, the aggregated data had a considerable increase of data points within each box, as shown in Figure 9. This increase of data points leads to a more comprehensive representation of each object, resulting in fewer identity switches and a decrease in FP.

However, by comparing the results for aggregated data in Tables 3 and 4, there is not a significant difference between aggregating data from 2 samples compared to 4 samples. In fact, the results from 2 samples performed bet-

ter than the results from 4 samples, even though aggregating data from 4 samples provides more data, as illustrated in Figure 9. The increased number of samples does not always result in an improved performance. This is because each sample represents 0.5s, 4 consecutive samples corresponds to a 2s time-delay between the first and last sample, which can be significant in high-speed roads. Moreover, aggregating more samples amplifies the computational overhead. Therefore, it is important to find an optimal balance in data aggregation that improves the framework without introducing excessive time delays and computational cost.

As previously discussed, the radar sensor is less affected by harsh weather conditions. Throughout this thesis, various tests have been conducted under rainy conditions to demonstrate if radar data can enhance the performance of the Poly-MOT framework during these conditions.

The results in Table 4 shows that radar sensor fusion archives an similar AMOTA comparable to LiDAR-based performance, achieving a result of 68.9%. The category that showed much improvement was radar’s ability to predict pedestrian movement during rainy weather, which is crucial for ensuring road safety and for developing robust ADAS that perform optimally in diverse environmental conditions. This is because the pedestrian moves relatively slowly and radar is good at capturing small changes in velocity. Additionally, the motion of pedestrian tend to move direct from the radar or towards the radar, minimizing the angular inaccuracies which will further be discussed in Section 7.2.

The best result was achieved with aggregated data over two samples but did not outperform the LiDAR-only approach, thus indicating that the computational overhead when aggregating radar data is not justified. It is important to note that the nuScenes dataset used in this thesis only includes scenarios with moderate rain. The results presented in this work indicate that the radar sensor is not capable of improving the tracking performance on this particular dataset. However, with a dataset that includes severe weather conditions, such as heavy rain, fog or snow, the expected performance gap between LiDAR and radar will widen in favor of radar sensors. During such conditions, this thesis implementation could be able to improve the object tracking results.

## 7.2 Discussion of Specific Results

From Table 3 it is clear that when the velocity measurement was solely derived from radar measurement the model performed poorly. This result is similar to the result highlighted in Table 2 where no velocity measurement was used. This can be due to several reasons, one being the fact that there is not always radar data available for each detected object. Another significant factor is the impact of angle of movement. Radar sensor provides the most accurate velocity measurement when the tracked object moves directly towards or directly away from the radar beam. However, as the angle comes closer to  $90^\circ$  the error increases see Figure 14. At  $90^\circ$ /perpendicular to the radar beam and the velocity direction, the radar will not be able to detect the velocity resulting in the measurement being zero. One method that has been proven effective in this paper and achieved improved results on object tracking is to use aggregated data. Collecting data from multiple radar data from different angles and position has helped to overcome issues related to angular movement. Additionally, 4D radar can address some of these issues. 4D radars offer adaptive beamforming techniques and higher resolution and therefore more radar points. These advancements enable the radar to adjust the shape and direction of the radar beam depending on the detected object and collect more data from different angles, thereby minimizing the effect of angular related measurement errors. Figure 15 highlights the contrast in the number of radar points per frame between regular 3D radar (nuScenes) and 4D radar.

In addition to using 4D radars, there are also different algorithms such as OBB (Oriented Bounding Box), L-fit, and EOBB (Extended Oriented Bounding Box) that can help with mitigating the impact of angular measurement errors. These algorithms offer various approaches to estimate the orientation and dimensions of objects more accurately [66]. Integrating these algorithms with 4D radar data could further enhance tracking performance by providing a more comprehensive understanding of object movement and improving robustness against angular inaccuracies.

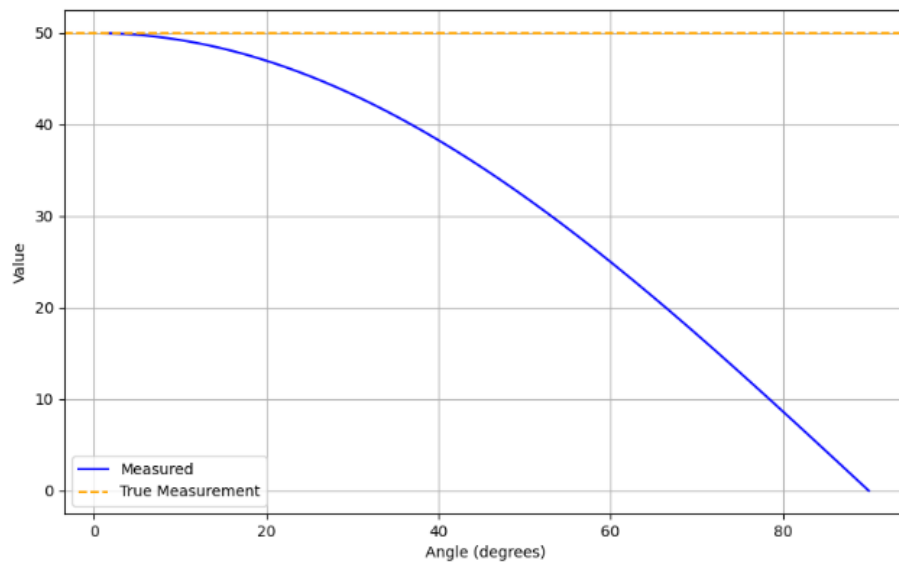


Figure 14: Radar sensor velocity deviation from real measured velocity from different angles between radar beam and object's movement



Figure 15: Radar PCD density. Difference visualization between 3D (top) and 4D (bottom) radar points [67].

### 7.3 Discussion of Method

This thesis has chosen to use mid level fusion to integrate radar velocity measurement. However, other fusion techniques could also be beneficial. Low-level fusion can be beneficial to fuse position data from LiDAR and radar to access its ability for long range detections. Although, implementing low-level fusion can result in longer computation time, since it processes a big amount of raw data from multiple radar and LiDAR points.

High-level sensor fusion is another fusion approach and can be useful. This fusion technique would handle preprocessed velocity measurement from LiDAR and radar, and could be quite beneficial in minimizing the impact of angular movement error radar suffers from, as previously mentioned in this thesis.

## 8 Conclusion

To conclude this thesis, the implemented method on the nuScenes dataset did not yield an improved result. This can be attributed to several factors. Firstly, the dataset does not include harsh weather conditions such as fog, heavy rain and snow that significantly affects the LiDAR sensor. These weather conditions are extremely valuable when evaluating the robustness of object tracking systems. Secondly, the available Radar points are sparse and limited for each sample. For a more effective object tracking system a comprehensive radar data is needed which nuScenes lacked.

Although a better object tracking result was not achieved, the thesis is still able to present the improvements of aggregate radar data compared to non aggregate radar data. By combining data from multiple timestep the model managed to create a more reliable and accurate velocity measurement.

The primary objective of this thesis was to evaluate the radar sensor's performance through sensor fusion. This work has presented the capabilities, struggles and limitation of radar which can be further investigated with 4D radar sensor and in harsh weather conditions.



## 9 Future Work

As demonstrated in the results, sensor fusion between LiDAR and radar was able to perform comparably to the Lidar-based framework Poly-MOT. This thesis aims to improve the performance of the framework by integrating radar data, this was not possible during this thesis. To accomplish this, there are several development opportunities for this work.

One approach is to reduce the redundancy of radar points by implementing refined filtering and pre-processing techniques. This reduction would decrease the computational power required to process the information and increase the quality of the data measurements. Another potential improvement involves addressing the issue of increased delay induced by aggregating data over multiple samples, thereby reducing the model's accuracy. A promising solution to this problem is the implementation of *Weighted Aggregation*. This technique assigns different weights to data points depending on their relevance to the task. In this case, more recent radar data points would be given higher weights compared to older ones, reducing the relevance of older data as delay increases. This approach would ensure the maintenance of accuracy while minimizing the impact of delayed data.

In addition to optimizing data aggregation techniques, it is essential to consider the performance of the sensor fusion framework under different conditions. The best evaluation performance for the sensor fusion framework was demonstrated during the “Weather-split” evaluation. It is expected that the radar would perform better during harsh weather conditions, a scenario we were unable to evaluate during this thesis. For future work, it would be valuable to evaluate the framework's performance during heavy rain or snow conditions. This evaluation would provide comprehensive information of how adverse weather affects the tracking performance and how well sensor fusion enhances robustness in such conditions.

Another approach is to implement a class specific object for the Kalman filter's measurement noise covariance matrix. For example, if the model is known to be less accurate in measuring the velocity of certain objects, such as motorcycles, the noise covariance should be set higher. Motorcycles, has a smaller size and erratic movement patterns compared to cars and pedestrians, present a greater challenge for accurate velocity measurement. By implementing dynamic noise covariance matrices for different class objects,

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the Kalman filter could make better predictions, leading to improved tracking performance.

Lastly, the framework should be applied to datasets from newer, state-of-the-art sensors to assess its adaptability and performance with the latest technology. Advances in sensor technology could offer higher resolution and more precise measurements, potentially enhancing the framework's accuracy and reliability.

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